Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	CASE-df001-2018-ios
Operator:	Floris
Analysis Started:	2026-06-15 15:16:07
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\train_phantom_3\df001\2018\ios
Output Directory:	C:\Users\Floris\Documents\dji_drones\train_phantom_3\df001\2018\dfdof_output\CASE-df001-2018-ios
Evidence Sources:	2
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner.The populated case output directory is the authorative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
flight_logs.zip	drone_flight_storage	logical	869.5 MB	2026-06-15 15:16:10
ios_logical.zip	controller_ios	logical	152.6 MB	2026-06-15 15:16:10

Evidence Source Path

C:\Users\Floris\Documents\dji_drones\train_phantom_3\df001\2018\ios

Cryptographic Hashes

flight_logs.zip	drone_flight_storage	logical
SHA-256:	`b0c00abba0c2deaddf3f70acf0e03b1b55b500c526b390b9d67de854e5d8b8ec`	
SHA-1:	`96a872d37a8b11f0d0ed78bb2da19431cdaf829b`	

ios_logical.zip	controller_ios	logical
SHA-256:	`e54eea13bcc93a8657925b590a141a801ef1925c06d395663cadf7d88fe6e6d4`	
SHA-1:	`a81d6af714e470a0dd910be851393c9c2894fbb9`	

2. Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p2 - controller android]: source evidence not found
2. [p3 - controller android]: source evidence not found
3. [p3 - controller ios - drone logs]: no artefacts found
4. [p3 - controller ios - images]: no artefacts found
5. [p3 - controller ios - videos]: no artefacts found
6. [p3 - drone sd]: source evidence not found
7. [p4 - controller android]: source evidence not found
8. [p4 - drone sd]: source evidence not found
9. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY000.csv
10. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY001.csv
11. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY002.csv
12. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY003.csv
13. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY004.csv
14. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY014.csv

Tool Execution Status

Tool	Invocations	Successes	Status
datcon	22	22	[OK]
parser_ios	1	1	[OK]
txtlogtocsvtool	1	1	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	2 / 4
Artefact Categories With Data	5 / 7
Flights With Primary Correlation	1 / 16
Tools Succeeded	3 / 3

3. Device and Account Identification

Field	Value	Confidence	Sources
Installed DJI App(s)	com.dji.go com.dji.pilot	[K] Controller-only	1
Controller Device	iPad mini 4	[K] Controller-only	1
Drone Serial Number	03Z1389682	[K] Controller-only	2
Drone Model	P3 Standard P3C P3S	[!] Inconsistent	18
Controller Device	iPad5,1__weibosdk__003143000__iphone__os 11.4	[K] Controller-only	2
DJI Application Version	3.1.38 4.2.20	[!] Inconsistent	3
Account E-Mail	droneforensics@vtoinc.com	[K] Controller-only	2
Product Type	2 7	[!] Inconsistent	2
Last Country Code	US	[K] Controller-only	2
Last App Launch	2018-06-19T18:58:13Z	[K] Controller-only	1
Last Flight Date	1970-01-01T00:00:00Z	[K] Controller-only	2

4. Flight Analysis - Flight 01

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `6330a8eedcd3f8ff7a6ba78b4355738d4056f98b4bec08831507aa2568c34c62`
Drone Log Name: FLY005.csv
Start: 2018-06-07T15:49:55.557000+00:00
End: 2018-06-07T15:54:24.160000+00:00
Duration: 268.6 s

Flight Metrics

Peak Height: -
Distance Travelled: -
Video Recording: No
Photo Capture: No

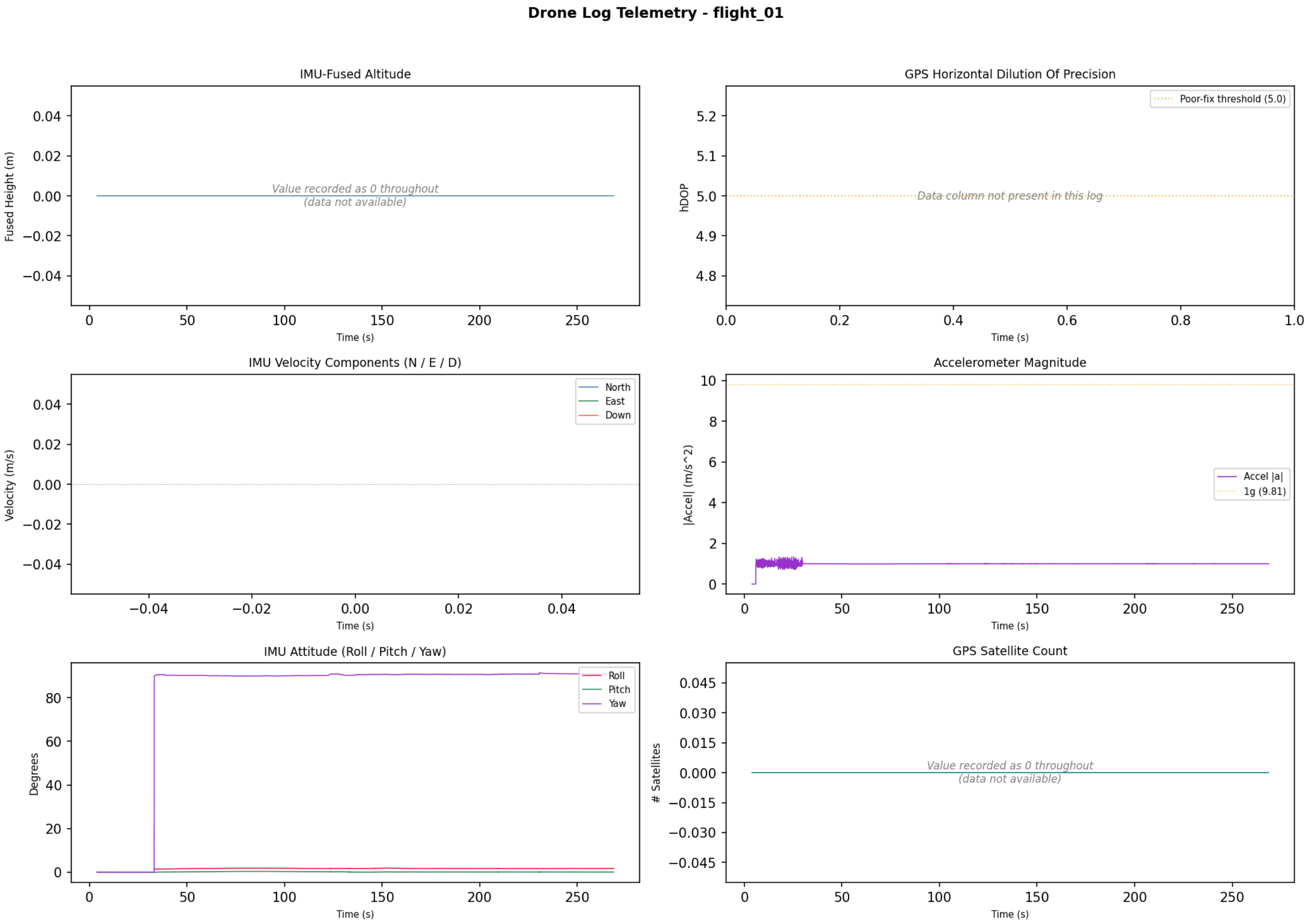
Fly Modes Observed (Chronological)

1. 0

Key Events

Timestamp (UTC)	Source	Event	Details
2018-06-07 15:50:47	drone_flight_storage:dl	Log started	P3S
2018-06-07 15:54:24	drone_flight_storage:dl	Log ended	
3236-01-12 23:59:48	drone_flight_storage:dl	Reached peak height	-0.0 m

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 02

Correlation and Flight Window

Status:	Unmatched (no drone log corroboration)
Drone Log SHA-256:	`65ca3eb3e94b3293c8627f682f92945867acbb5944751cbcc13c7d29659d8043`
Drone Log Name:	FLY006.csv
Start:	2018-06-07T16:41:32.117000+00:00
End:	2018-06-07T16:43:53.300000+00:00
Duration:	141.2 s

Flight Metrics

Peak Height:	-
Distance Travelled:	0.01228646116311485 m
Video Recording:	No
Photo Capture:	No

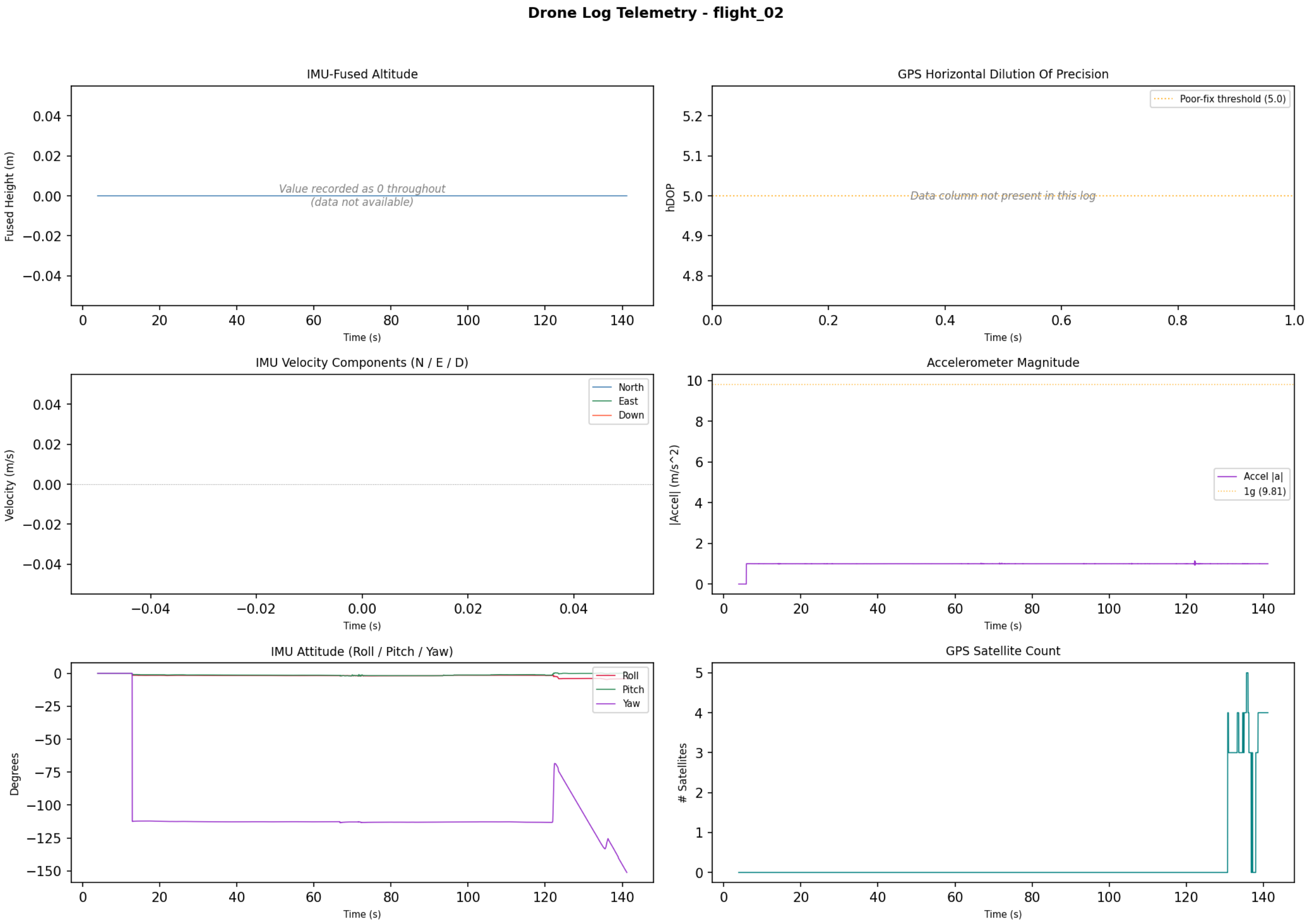
Fly Modes Observed (Chronological)

1. 0

Key Events

Timestamp (UTC)	Source	Event	Details
2018-06-07 16:41:36	drone_flight_storage:dl	Log started	P3S
2018-06-07 16:41:36	drone_flight_storage:dl	Reached peak height	-0.0 m
2018-06-07 16:43:56	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 03

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `ae702aa8ef96dbfe2763f478fe48ebfefff50092f1a3e0976cb4861899e0edcd`
Drone Log Name: FLY007.csv
Start: 2018-06-07T16:48:35.118000+00:00
End: 2018-06-07T16:50:51.300000+00:00
Duration: 136.2 s

Flight Metrics

Peak Height: -
Distance Travelled: 58.783664825212966 m
Video Recording: No
Photo Capture: No

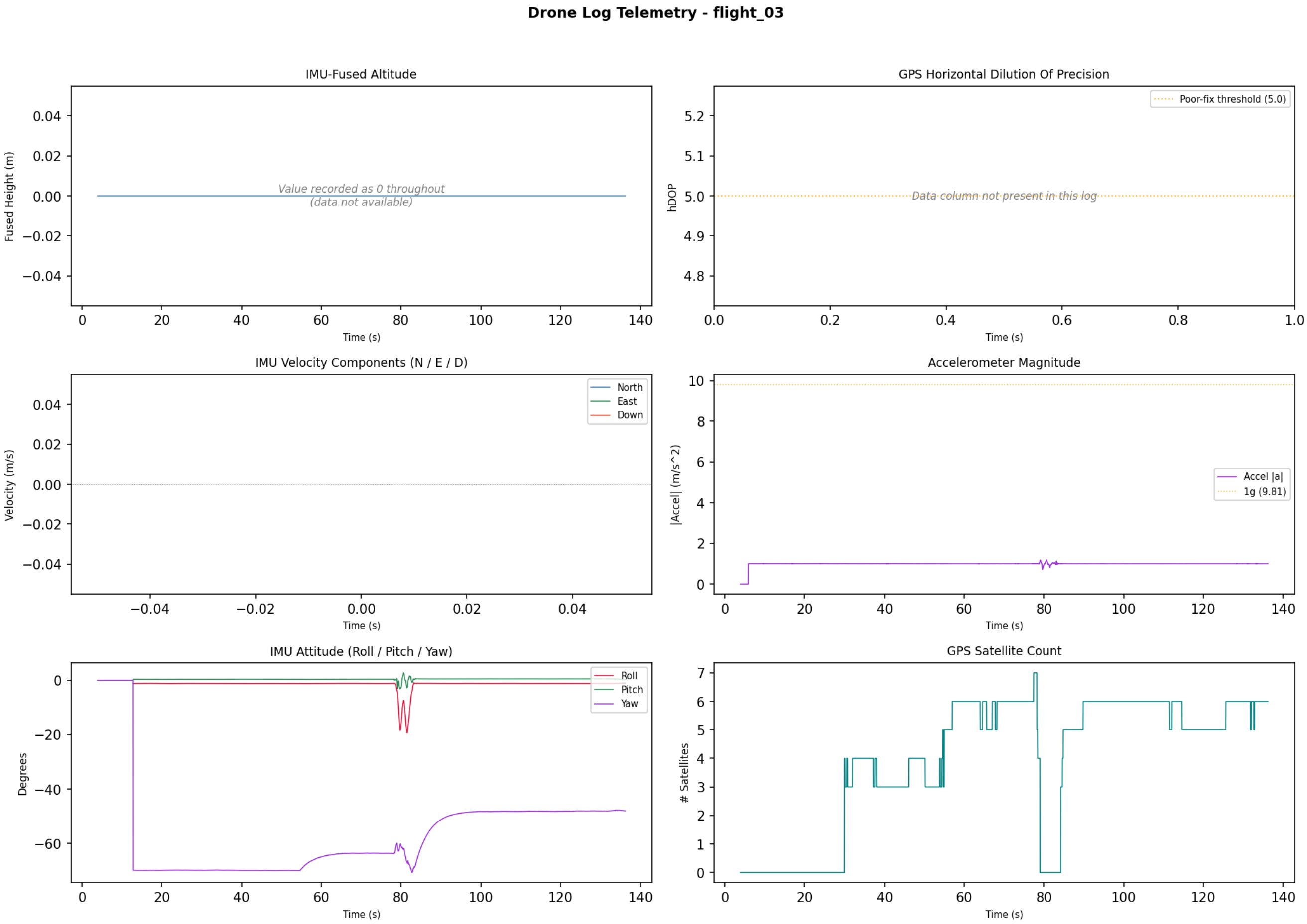
Fly Modes Observed (Chronological)

1. 0

Key Events

Timestamp (UTC)	Source	Event	Details
2018-06-07 16:48:39	drone_flight_storage:dl	Log started	P3S
2018-06-07 16:48:39	drone_flight_storage:dl	Reached peak height	-0.0 m
2018-06-07 16:50:54	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 04

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `a27ad6b90d8e3d2ce3f4715a93b866948df4f66ec7faf8fa74a14ab21fc994ff`
Drone Log Name: FLY009.csv
Start: 2018-06-16T23:59:46.463000+00:00
End: 2018-06-17T00:05:48.015000+00:00
Duration: 361.6 s

Flight Metrics

Peak Height: 0.09117607116368376 m
Distance Travelled: 4.0208613717078165 m
Video Recording: No
Photo Capture: No

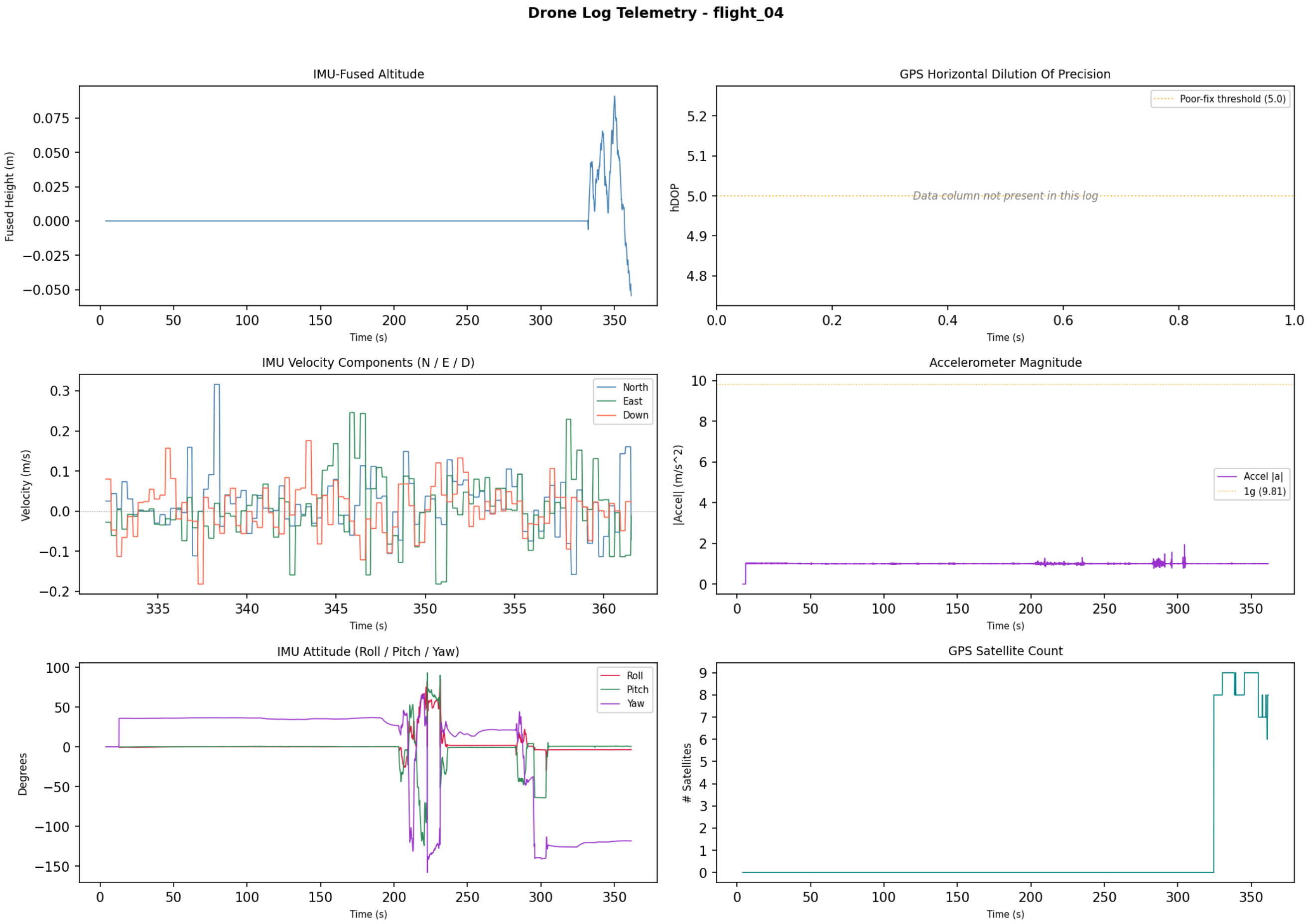
Fly Modes Observed (Chronological)

1. 0

Key Events

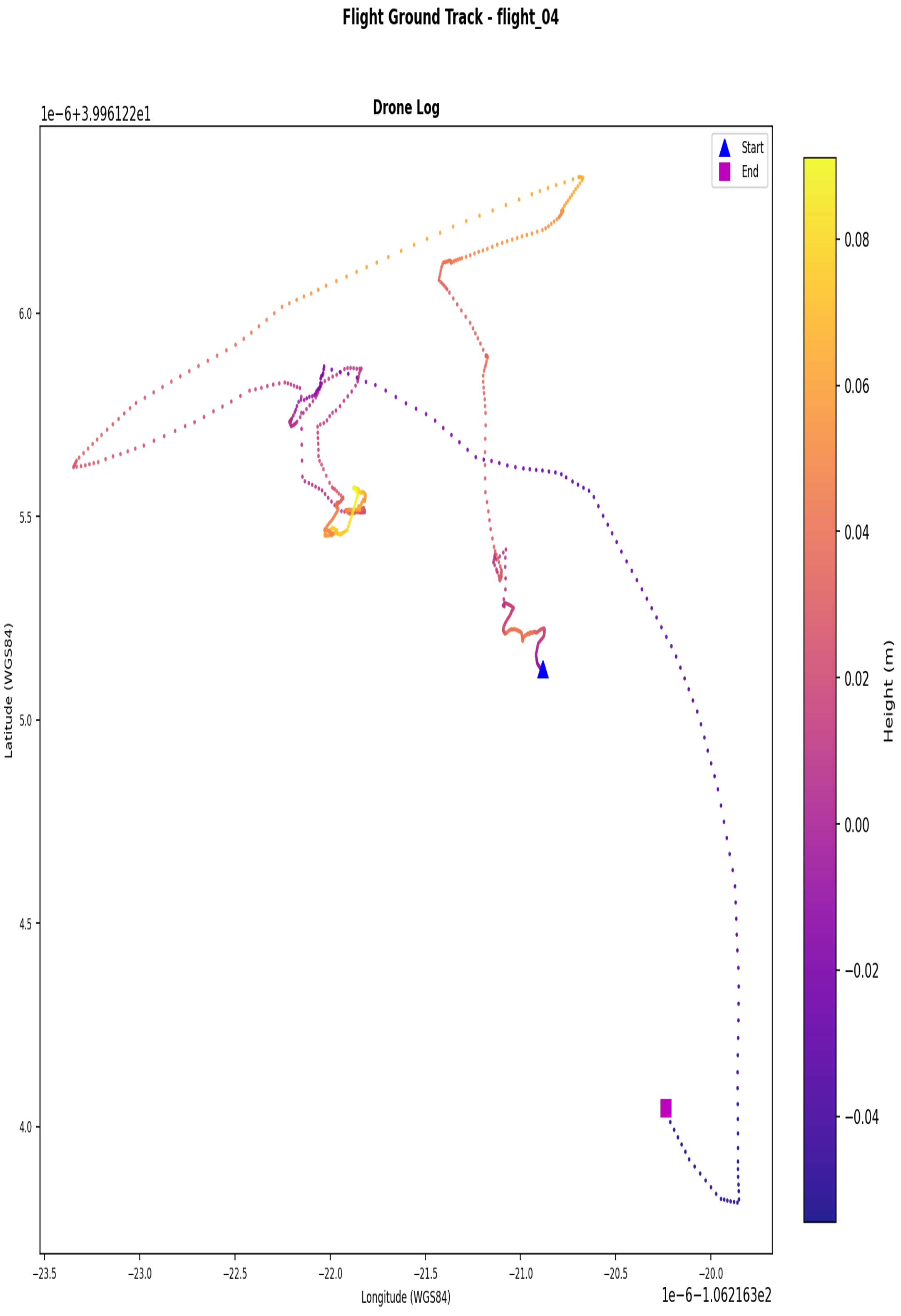
Timestamp (UTC)	Source	Event	Details
2018-06-17 00:04:36	drone_flight_storage:dl	Log started	P3S
2018-06-20 14:21:22	drone_flight_storage:dl	Reached peak height	0.1 m
2018-06-20 14:21:34	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY009.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).
A KML file (FLY009.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 05

Correlation and Flight Window

Status:	Matched (primary GPS+temporal)
Confidence:	medium
GPS Overlap:	344.9 s
Median GPS Distance:	6.79 m
Overlap:	100%
GPS Match Rate:	100%
Flight Record SHA-256:	820a98408bf6b140b572568cb5b1d14de0d59da6388d69fa26f2c9422a5d9b6f
Flight Record Name:	DJIFlightRecord_2018-06-19__13-16-58_.csv
Start:	2018-06-19T19:16:58.134000+00:00
End:	2018-06-19T19:22:43.046000+00:00
Duration:	344.9 s
Drone Log SHA-256:	72f271c20306ff986e07eb0a8e5de564dd77339a67955cc17146c06d43ed6eb2
Drone Log Name:	FLY008.csv
Start:	2018-06-19T18:59:03.770000+00:00
End:	2018-06-19T19:23:38.760000+00:00
Duration:	1475.0 s

Flight Metrics

Peak Height:	28.8 m
Distance Travelled:	1679.22 m
Video Recording:	Yes
Photo Capture:	No
Photos Taken:	0
Recording Duration:	156.0 s
Record Complete:	Yes

Fly Modes Observed (Chronological)

1. 1
2. 2
3. GPS_Atti
4. AutoTakeoff
5. Atti
6. AutoLanding
7. 3
8. 0

Key Events

Timestamp (UTC)	Source	Event	Details
2018-06-19 18:59:17	drone_flight_storage:dl	Log started	P3S
2018-06-19 19:16:58	ctrl_ios:fr (v3.1.38)	Log started	SN: 03Z1389682; App: 3.1.38; P3 Standard
2018-06-19 19:16:58	ctrl_ios:fr (v3.1.38)	Motor turned on	on
2018-06-19 19:16:58	ctrl_ios:fr (v3.1.38)	Record mode changed	No
2018-06-19 19:19:35	ctrl_ios:fr (v3.1.38)	Reached peak height	28.8 m
2018-06-19 19:20:06	ctrl_ios:fr (v3.1.38)	Record mode changed	Starting
2018-06-19 19:20:30	drone_flight_storage:dl	Reached peak height	39.5 m
2018-06-19 19:22:43	ctrl_ios:fr (v3.1.38)	Log ended	Dist home: 0.8 m; Height: 0.0 m; Rec: 156.0 s
2018-06-19 19:22:43	ctrl_ios:fr (v3.1.38)	Motor turned off	off
2018-06-19 19:23:35	drone_flight_storage:dl	Log ended	

In-Flight Warnings

Timestamp (UTC)	Source	Warning Message

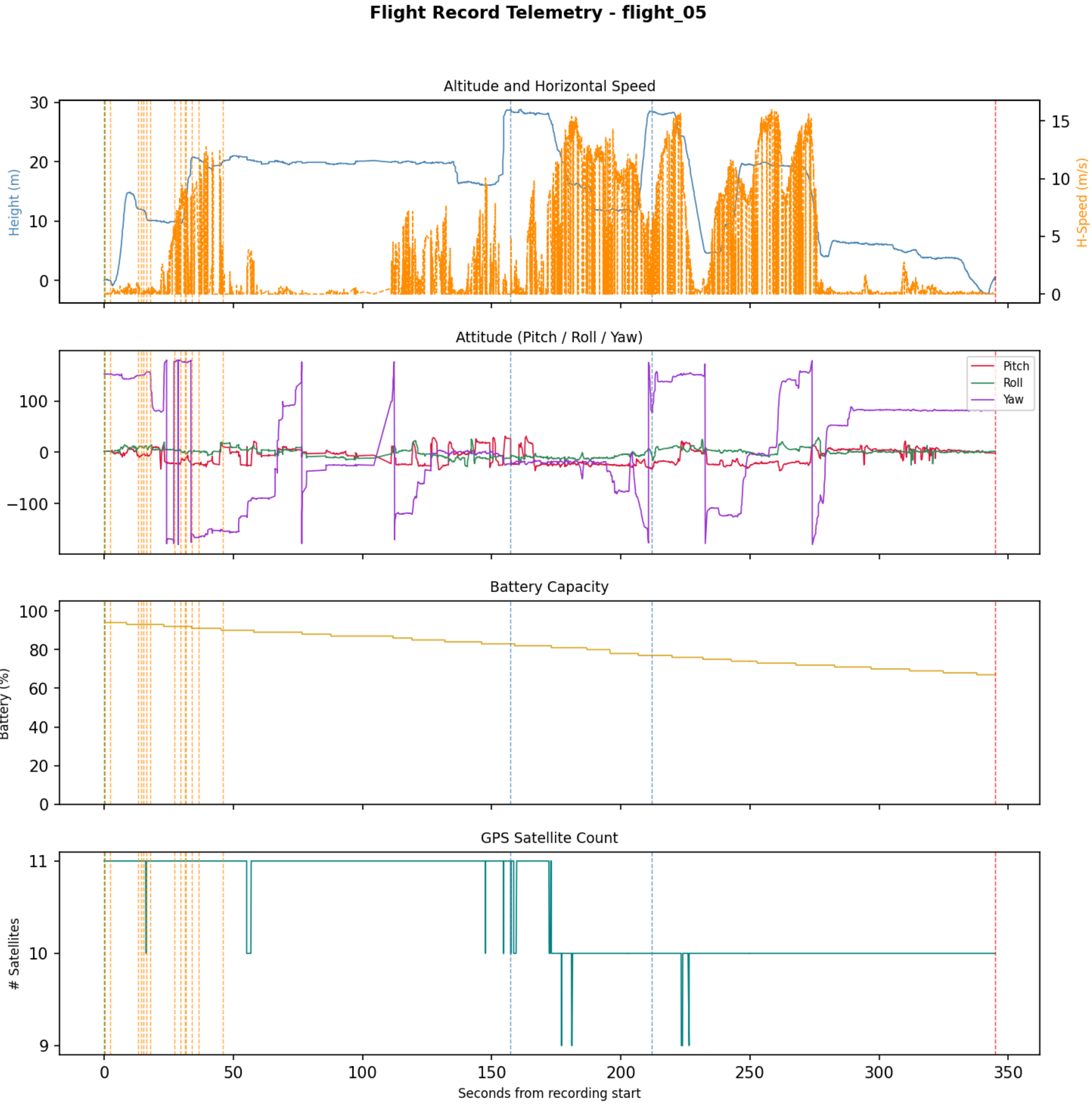
Timestamp (UTC)	Source	Warning Message
2018-06-19 19:16:58	ctrl_ios:fr (v3.1.38)	Warning:Strong Interference Detected. Be careful when flying long distances.
2018-06-19 19:17:00	ctrl_ios:fr (v3.1.38)	Warning:Strong Interference Detected. Be careful when flying long distances.
2018-06-19 19:17:11	ctrl_ios:fr (v3.1.38)	Warning:Strong Interference Detected. Be careful when flying long distances.
2018-06-19 19:17:12	ctrl_ios:fr (v3.1.38)	Warning:Strong Interference Detected. Be careful when flying long distances.
2018-06-19 19:17:13	ctrl_ios:fr (v3.1.38)	Warning:Strong Interference Detected. Be careful when flying long distances.
2018-06-19 19:17:14	ctrl_ios:fr (v3.1.38)	Warning:Strong Interference Detected. Be careful when flying long distances.
2018-06-19 19:17:15	ctrl_ios:fr (v3.1.38)	Warning:Strong Interference Detected. Be careful when flying long distances.
2018-06-19 19:17:25	ctrl_ios:fr (v3.1.38)	Warning:Strong Interference Detected. Be careful when flying long distances.
2018-06-19 19:17:27	ctrl_ios:fr (v3.1.38)	Warning:Strong Interference Detected. Be careful when flying long distances.
2018-06-19 19:17:29	ctrl_ios:fr (v3.1.38)	Warning:Compass Error. Exit P-GPS Mode
2018-06-19 19:17:29	ctrl_ios:fr (v3.1.38)	Warning:Strong Interference Detected. Be careful when flying long distances.
2018-06-19 19:17:32	ctrl_ios:fr (v3.1.38)	Warning:Strong Interference Detected. Be careful when flying long distances.
2018-06-19 19:17:34	ctrl_ios:fr (v3.1.38)	Warning:Strong Interference Detected. Be careful when flying long distances.
2018-06-19 19:17:44	ctrl_ios:fr (v3.1.38)	Warning:Strong Interference Detected. Be careful when flying long distances.

Correlated Artefacts

Possibly correlated (indirect indicators):

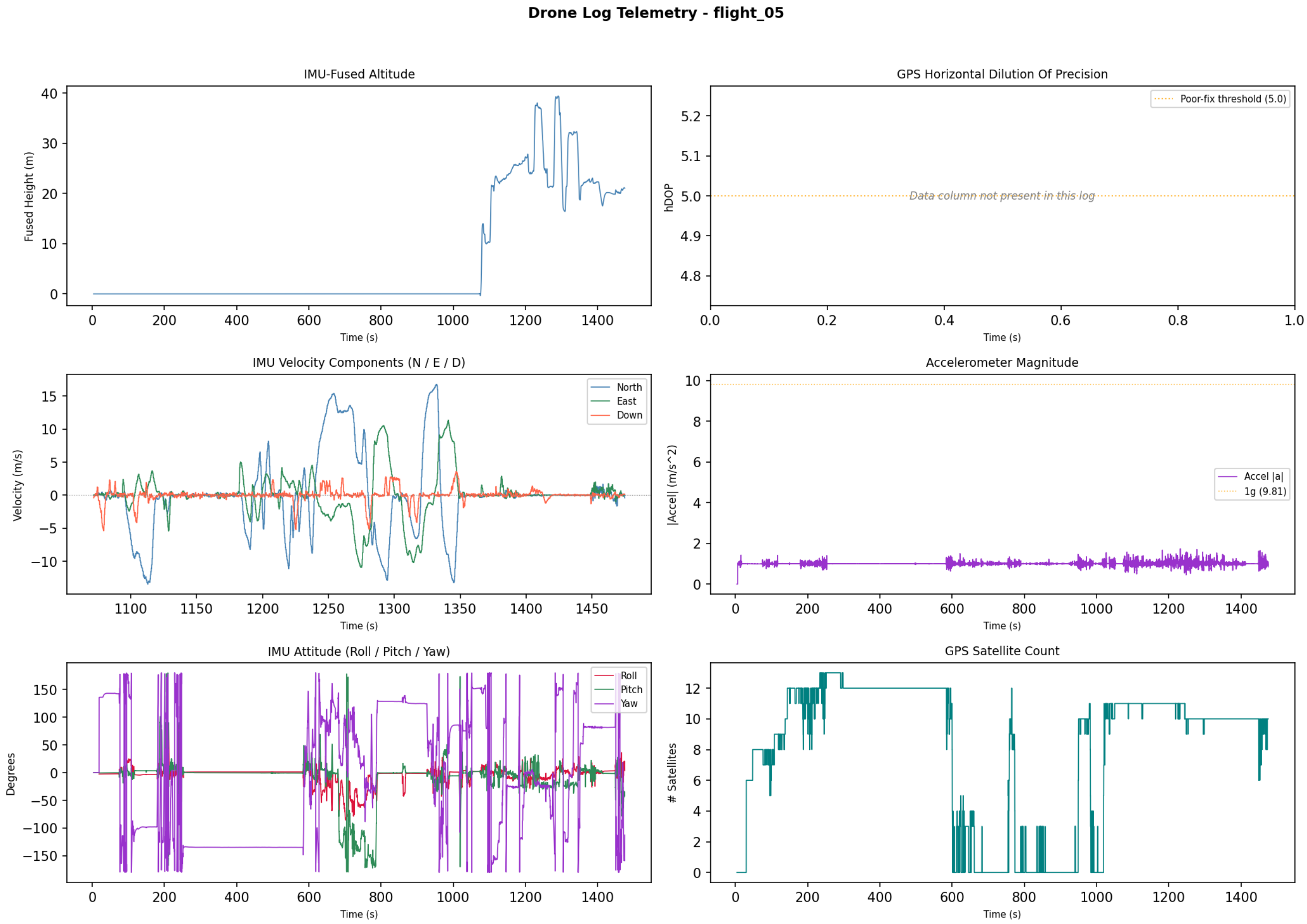
controller_ios: flight_logs	Log messages match (1 entries, format: json_array_log)
p5:3b54b251405cd0ce4832ee6e7efb646837e6f9748c2167e8b0986a9582ef1ebd	

Flight Record - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Drone Log - Sensor Telemetry



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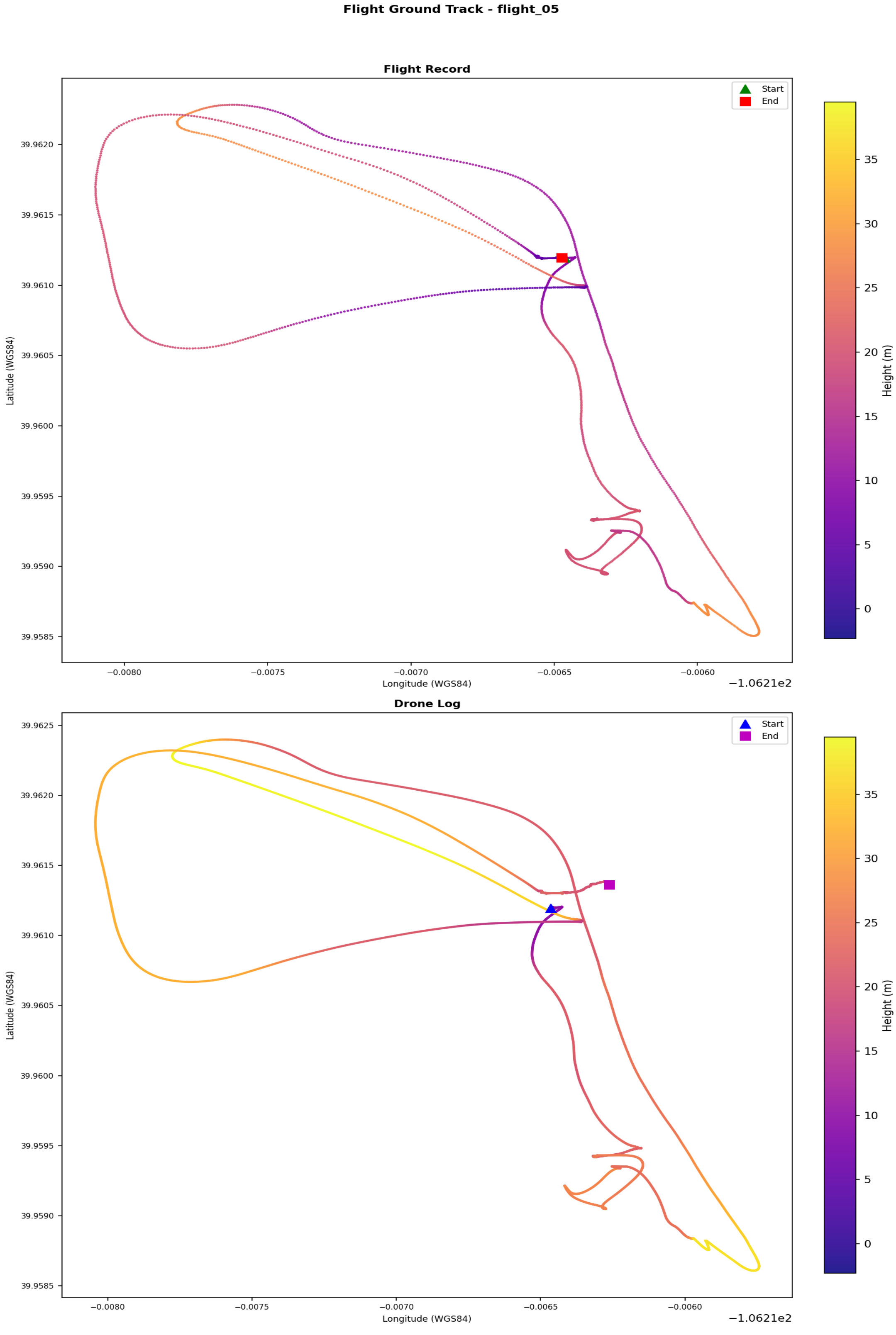
Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY008.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY008.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 06

Correlation and Flight Window

Status:	Unmatched (no drone log corroboration)
Drone Log SHA-256:	`78fbdc9fab8a9fb5c2eeeebb7ce94576158c456fcda5520ab06fb9206d9f5af1`
Drone Log Name:	FLY010.csv
Start:	2018-06-20T14:40:43.103000+00:00
End:	2018-06-20T14:40:54.105000+00:00
Duration:	11.0 s

Flight Metrics

Peak Height:	-
Distance Travelled:	-
Video Recording:	No
Photo Capture:	No

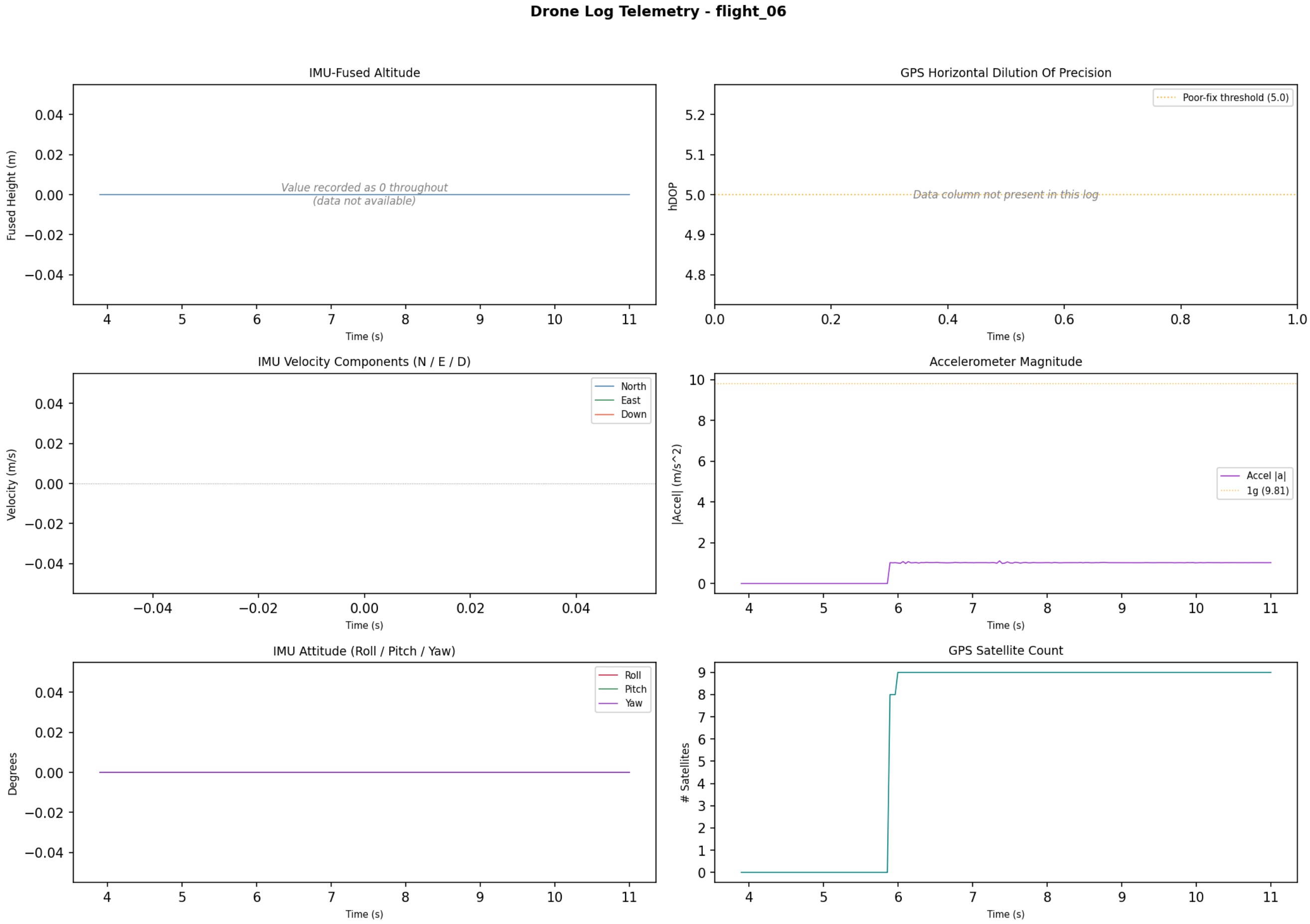
Fly Modes Observed (Chronological)

1. 0

Key Events

Timestamp (UTC)	Source	Event	Details
2018-06-20 14:40:47	drone_flight_storage:dl	Log started	P3S
2018-06-20 14:40:47	drone_flight_storage:dl	Reached peak height	-0.0 m
2018-06-20 14:40:56	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 07

Correlation and Flight Window

Status:	Unmatched (no drone log corroboration)
Drone Log SHA-256:	`bdff755e0f19d86a907baefc21a0ccb0bf6c1df389c378589c2853f5b3c3be0f`
Drone Log Name:	FLY011.csv
Start:	2018-06-20T14:58:54.075000+00:00
End:	2018-06-20T15:00:26.280000+00:00
Duration:	92.2 s

Flight Metrics

Peak Height:	7.154927647096823 m
Distance Travelled:	11.007765231291225 m
Video Recording:	No
Photo Capture:	No

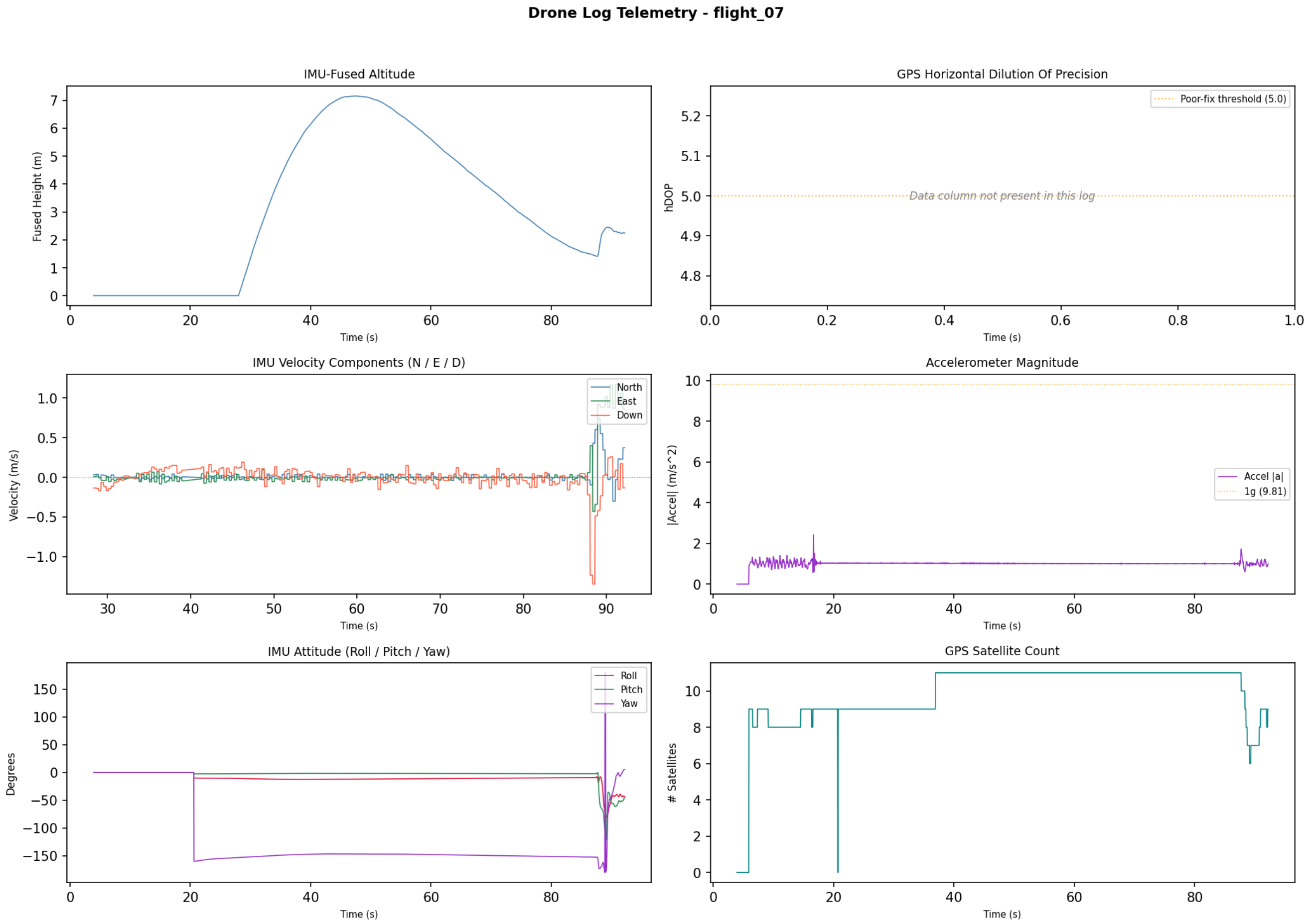
Fly Modes Observed (Chronological)

1. 0

Key Events

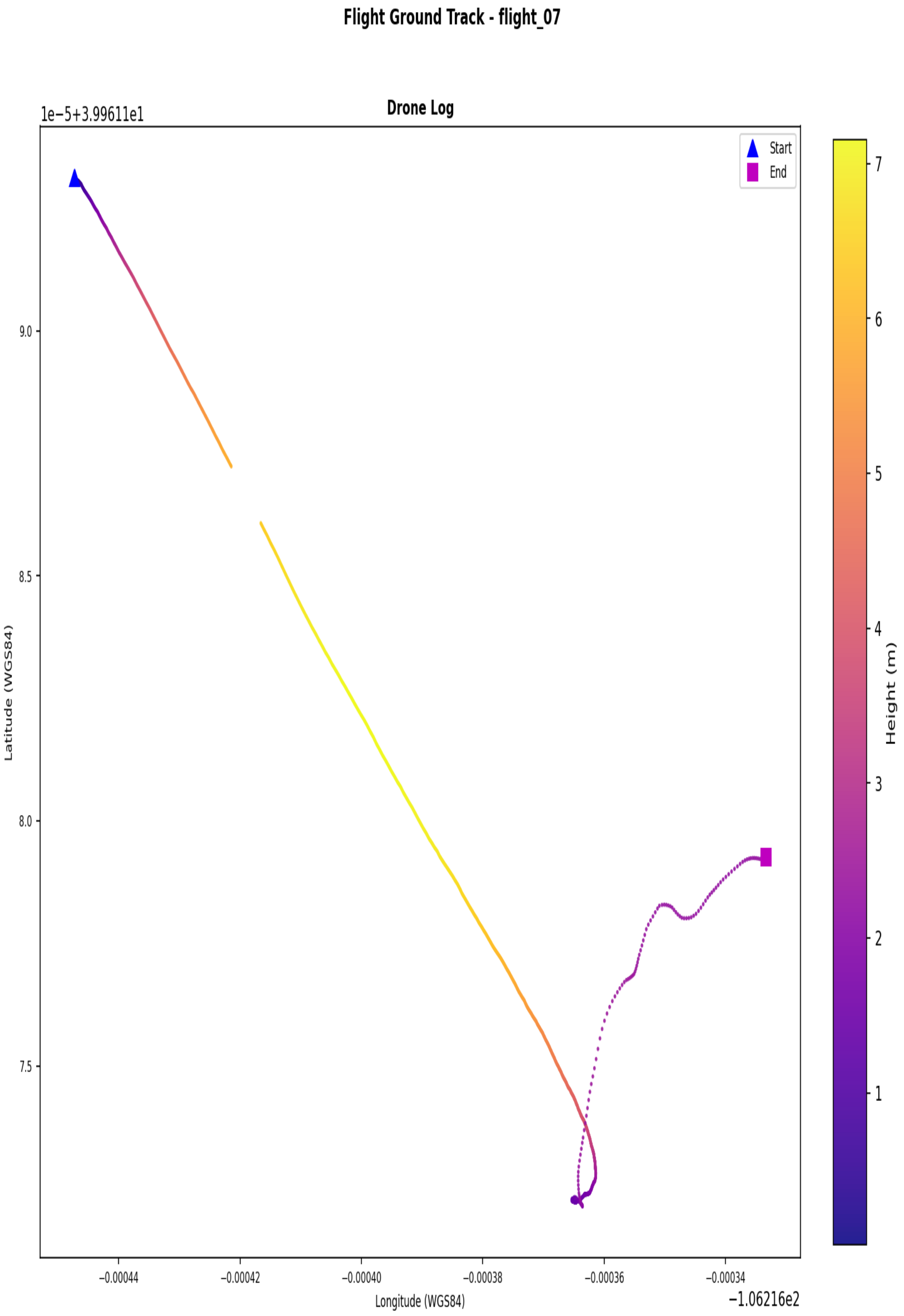
Timestamp (UTC)	Source	Event	Details
2018-06-20 14:58:58	drone_flight_storage:dl	Log started	P3S
2018-06-20 14:59:43	drone_flight_storage:dl	Reached peak height	7.2 m
2018-06-20 15:00:28	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY011.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).
A KML file (FLY011.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 08

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `c96fa8c2fb3bc8480e67f51c197f8c1d32c7c4953dae24d371e09b2823ddbc20`
Drone Log Name: FLY012.csv
Start: 2018-06-20T15:01:11.105000+00:00
End: 2018-06-20T15:03:13.800000+00:00
Duration: 122.7 s

Flight Metrics

Peak Height: 1.3026932382708851 m
Distance Travelled: 4.927617715712613 m
Video Recording: No
Photo Capture: No

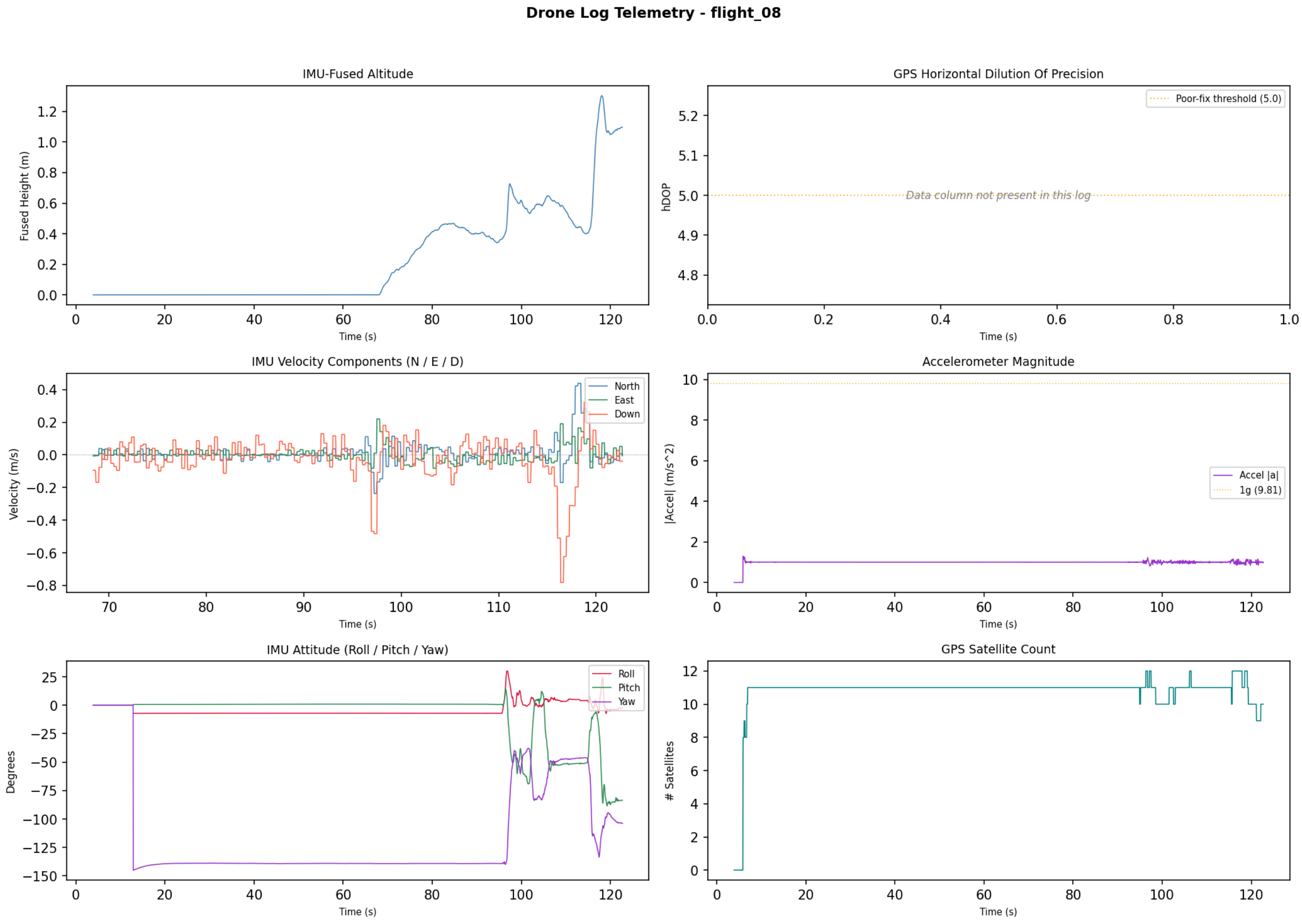
Fly Modes Observed (Chronological)

1. 0

Key Events

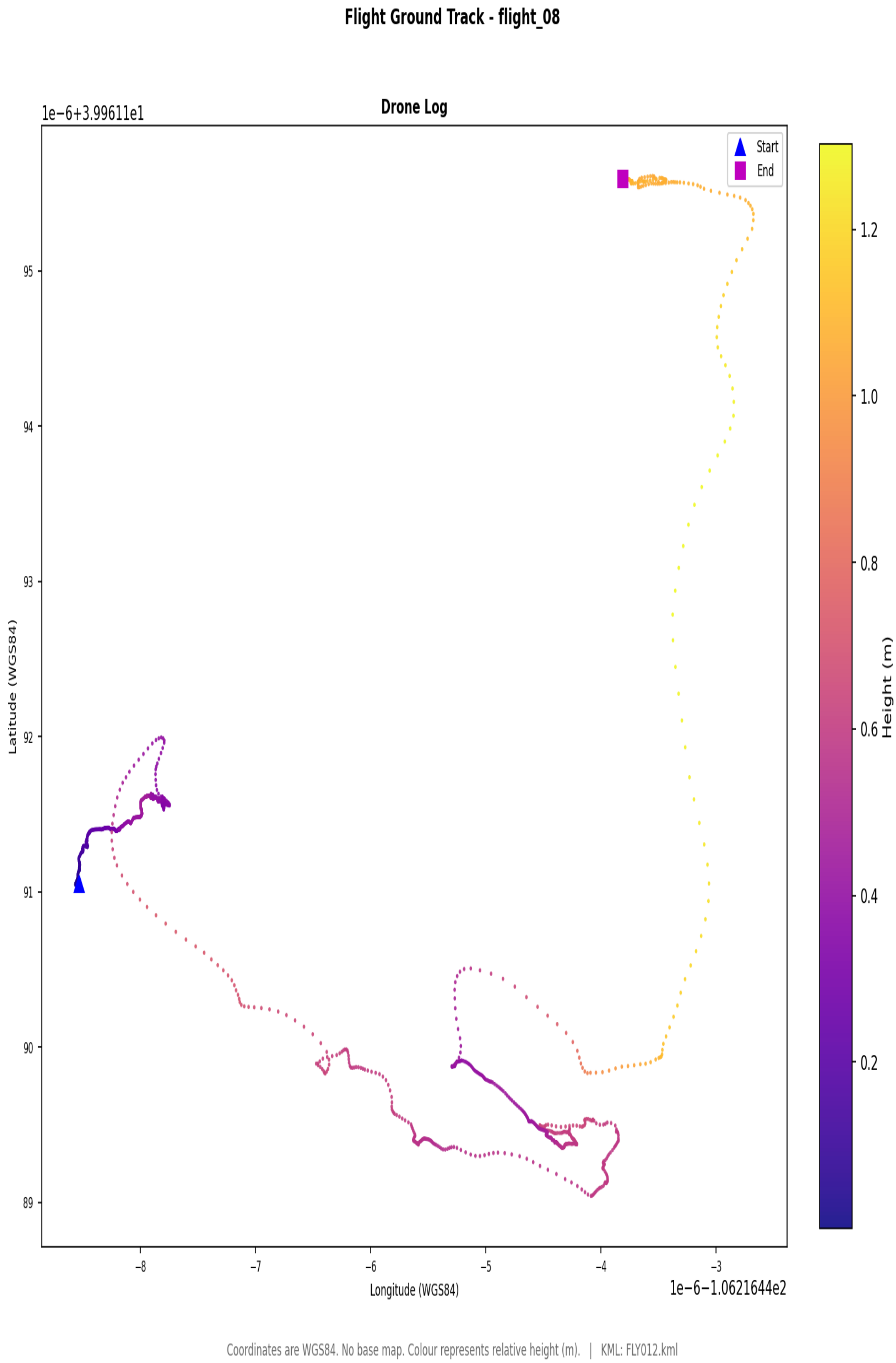
Timestamp (UTC)	Source	Event	Details
2018-06-20 15:01:15	drone_flight_storage:dl	Log started	P3S
2018-06-20 15:03:11	drone_flight_storage:dl	Reached peak height	1.3 m
2018-06-20 15:03:16	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY012.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 09

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `b4e674e670d3e40d779bf7b0856d21225202313e220e194b2a4d853bbef740ca`
Drone Log Name: FLY013.csv
Start: 2018-06-20T15:03:38.065000+00:00
End: 2018-06-20T15:08:35.560000+00:00
Duration: 297.5 s

Flight Metrics

Peak Height: 55.84417053056328 m
Distance Travelled: 1705.456865466296 m
Video Recording: No
Photo Capture: No

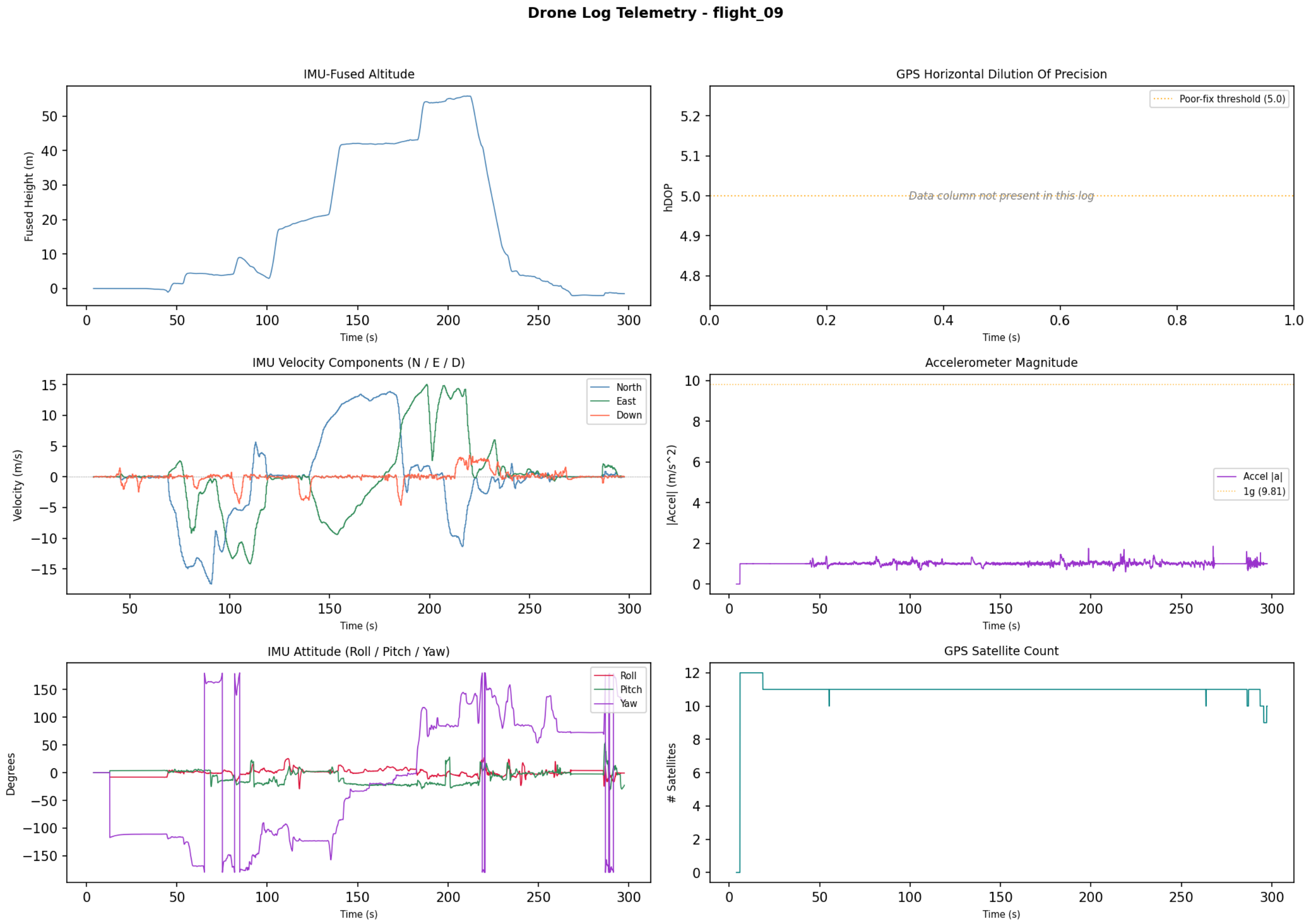
Fly Modes Observed (Chronological)

1. 0
2. 2
3. 1
4. 3

Key Events

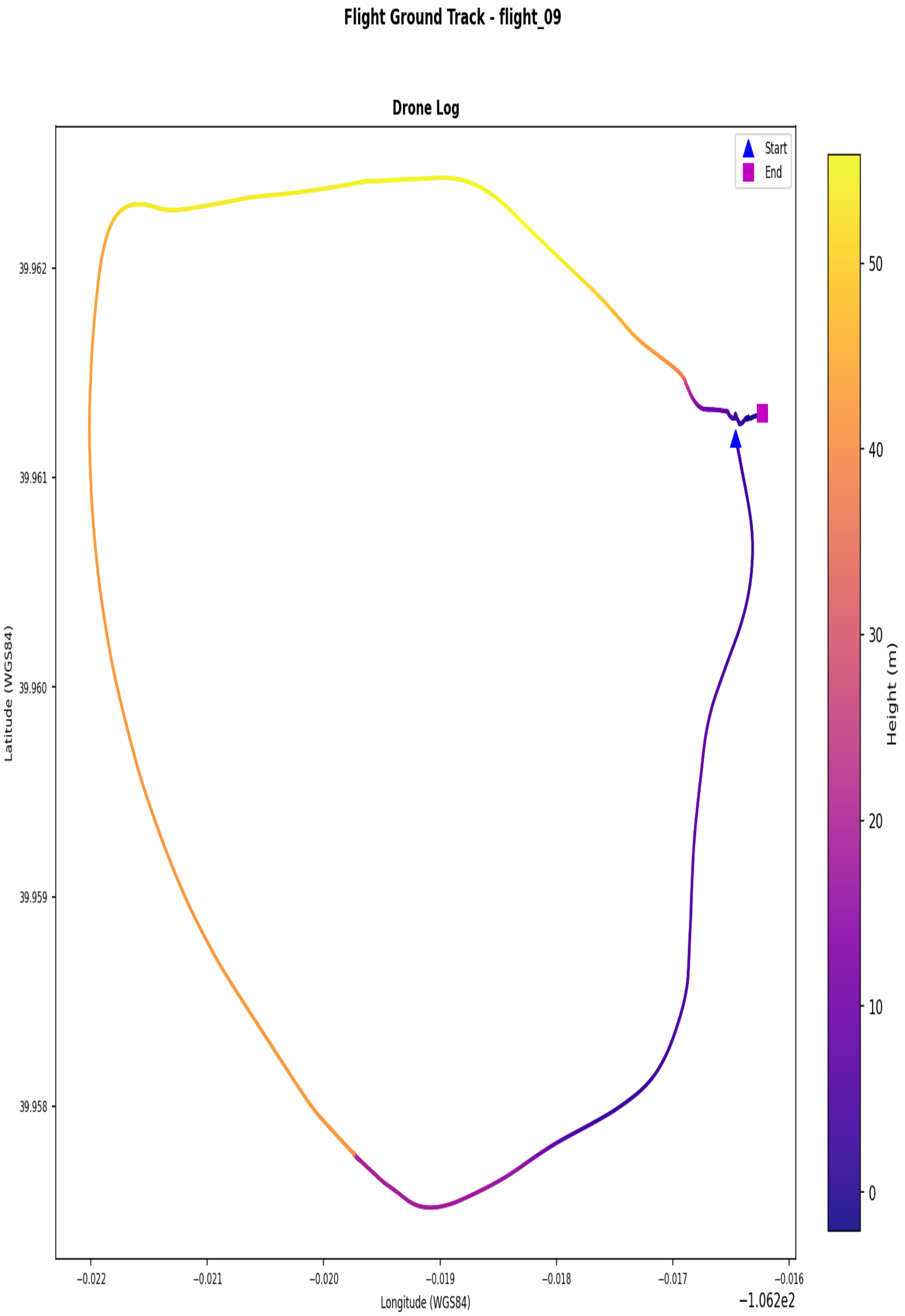
Timestamp (UTC)	Source	Event	Details
2018-06-20 15:03:42	drone_flight_storage:dl	Log started	P3S
2018-06-20 15:07:10	drone_flight_storage:dl	Reached peak height	55.8 m
2018-06-20 15:08:37	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY013.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).
A KML file (FLY013.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 10

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `34e6aa2645b4ec10935bcf1aa1efdd47cbcf9f3629c8d63b01e72d1f8dce6554`
Drone Log Name: FLY015.csv
Start: 2018-06-20T20:30:03.173000+00:00
End: 2018-06-20T20:39:23.185000+00:00
Duration: 560.0 s

Flight Metrics

Peak Height: 31.798286702144047 m
Distance Travelled: 775.5852552092165 m
Video Recording: No
Photo Capture: No

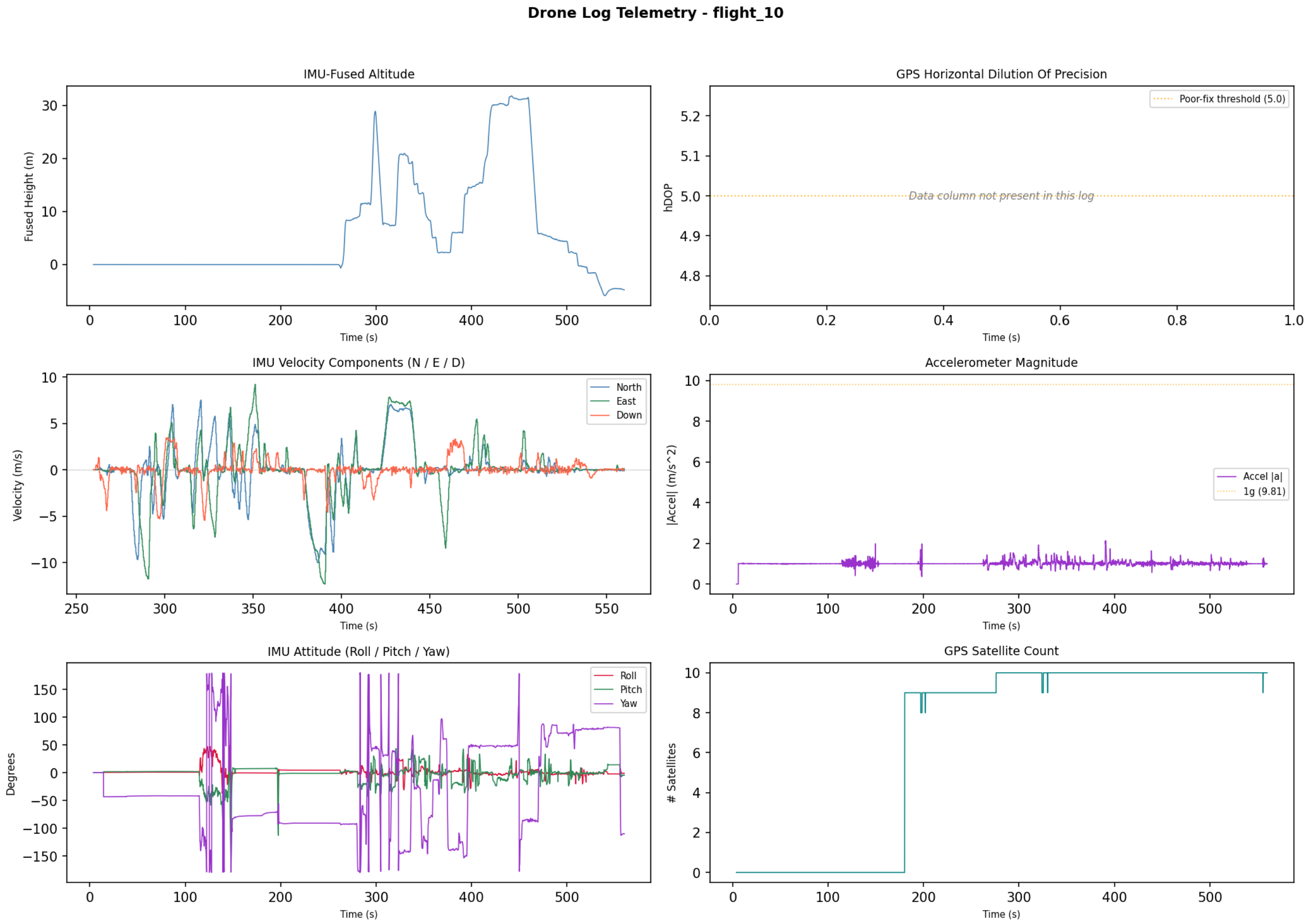
Fly Modes Observed (Chronological)

1. 2
2. 1
3. 3
4. 0

Key Events

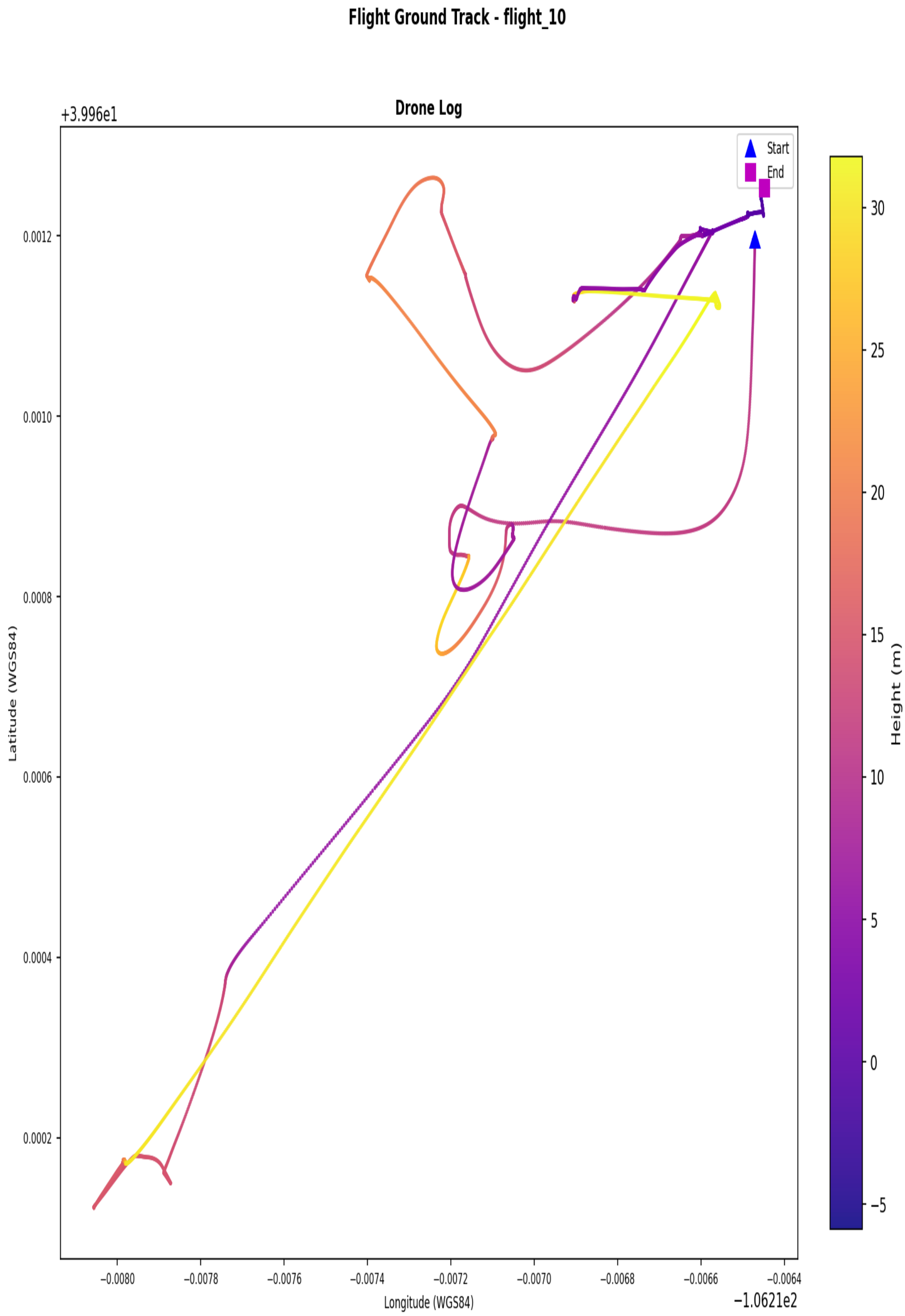
Timestamp (UTC)	Source	Event	Details
2018-06-20 20:32:17	drone_flight_storage:dl	Log started	P3S
2018-06-20 20:37:24	drone_flight_storage:dl	Reached peak height	31.8 m
2018-06-20 20:39:20	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY015.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).
A KML file (FLY015.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 11

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `4613ebd15162f415d0a68df86299496f34e43304bcf8651b0bb2b8f4e40c6ad5`
Drone Log Name: FLY016.csv
Start: 2018-06-21T14:48:30.252000+00:00
End: 2018-06-21T14:50:25.420000+00:00
Duration: 115.2 s

Flight Metrics

Peak Height: 1.9128186593879792 m
Distance Travelled: 7.320542195992152 m
Video Recording: No
Photo Capture: No

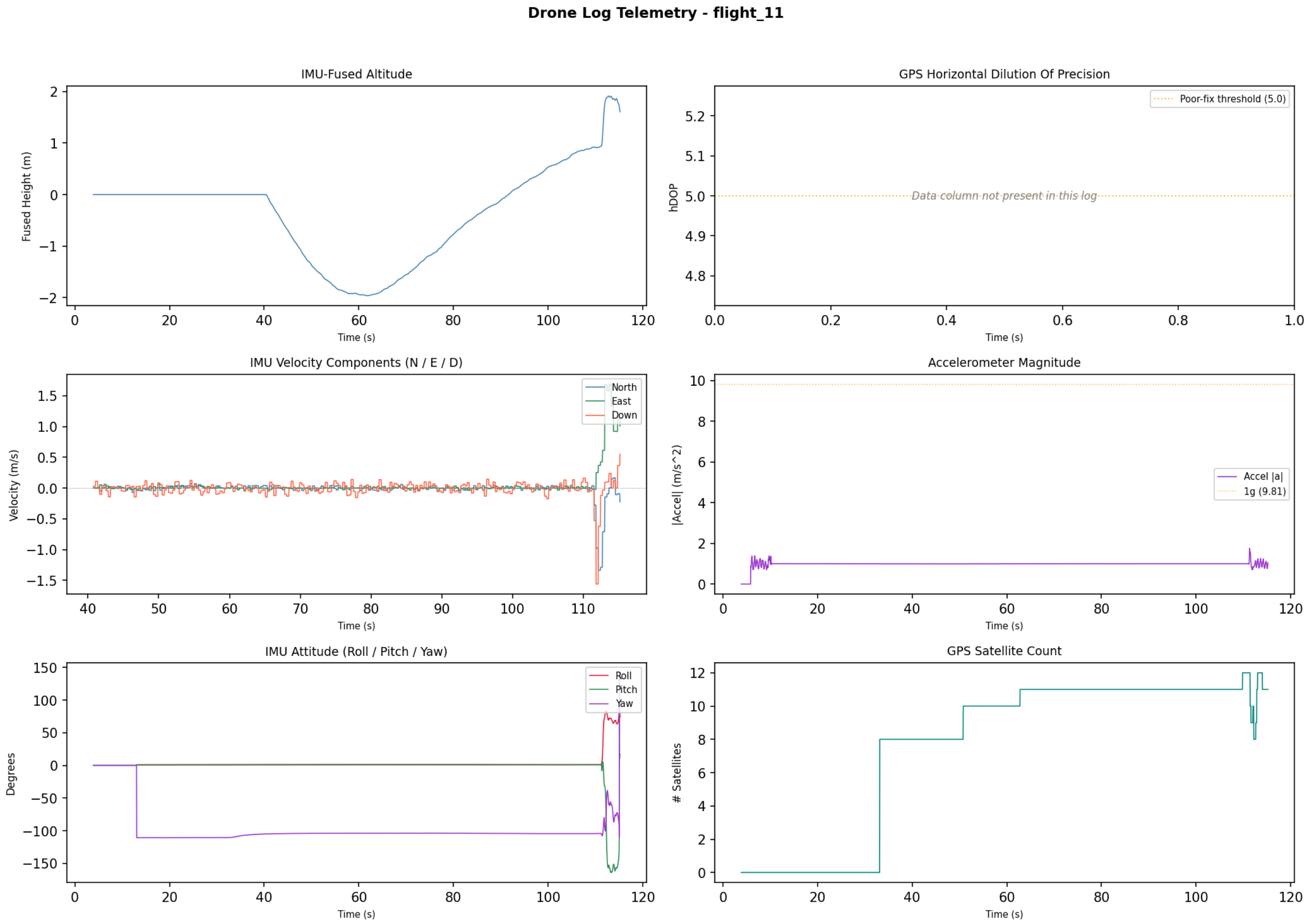
Fly Modes Observed (Chronological)

1. 0

Key Events

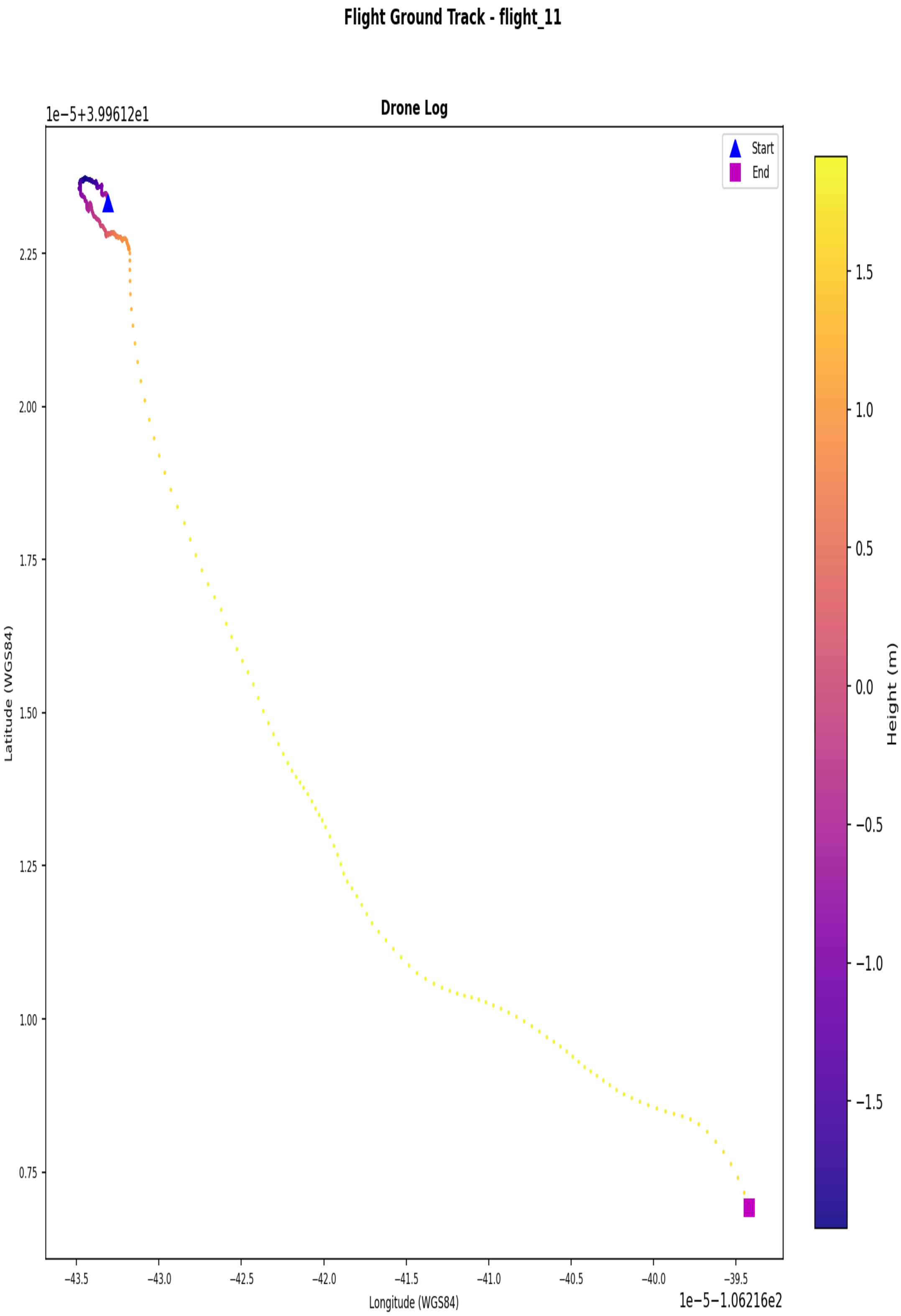
Timestamp (UTC)	Source	Event	Details
2018-06-21 14:48:47	drone_flight_storage:dl	Log started	P3S
2018-06-21 14:50:22	drone_flight_storage:dl	Reached peak height	1.9 m
2018-06-21 14:50:25	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY016.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY016.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 12

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `bfbb5b05396ef26b46229c5d5ca3ae235e5ae81589d2a96d8600ef6692aa2aad`
Drone Log Name: FLY017.csv
Start: 2018-06-21T15:51:40.662000+00:00
End: 2018-06-21T16:07:33.535000+00:00
Duration: 952.9 s

Flight Metrics

Peak Height: 107.54388886428954 m
Distance Travelled: 2933.0435180057566 m
Video Recording: No
Photo Capture: No

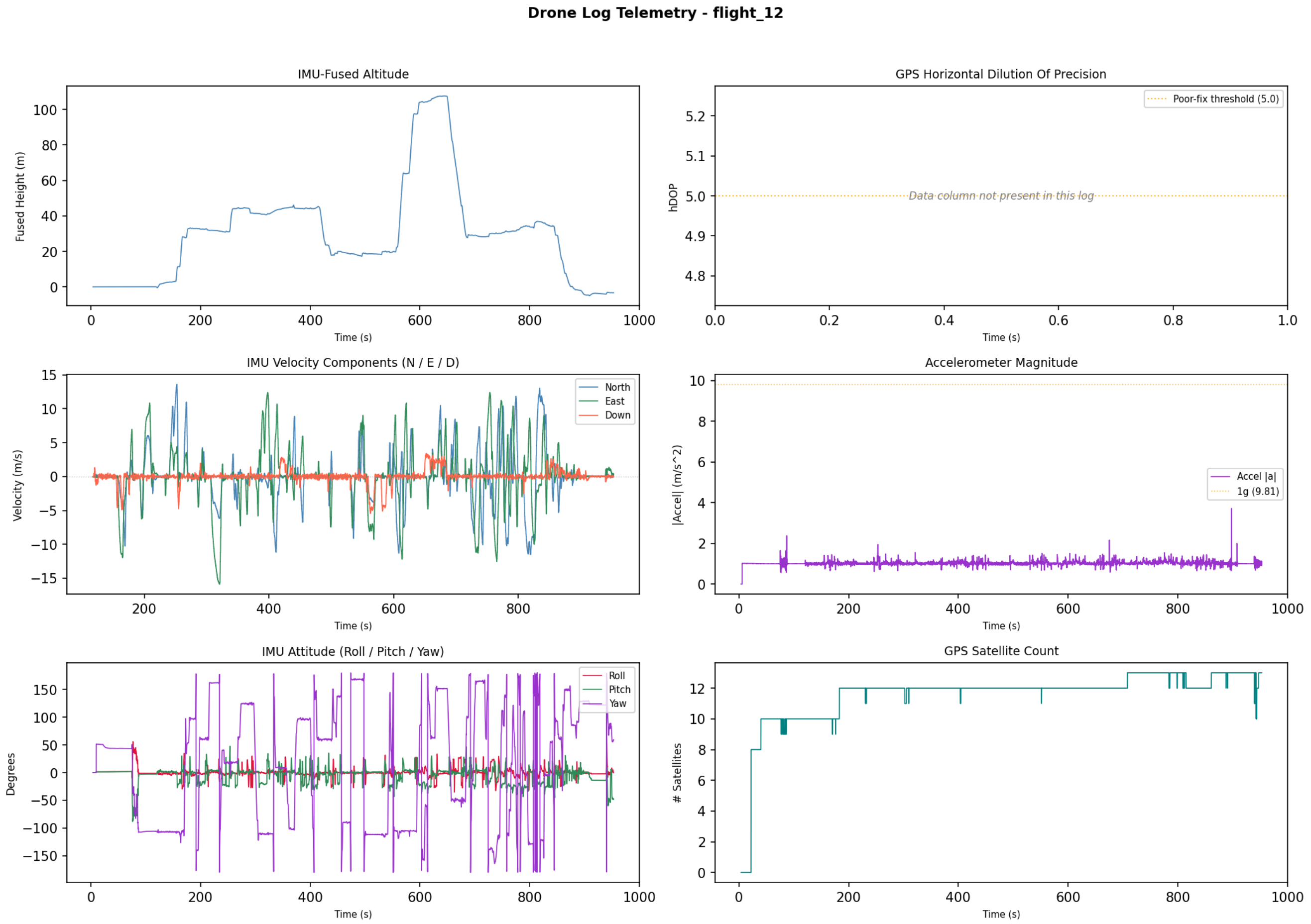
Fly Modes Observed (Chronological)

1. 2
2. 1
3. 3
4. 0

Key Events

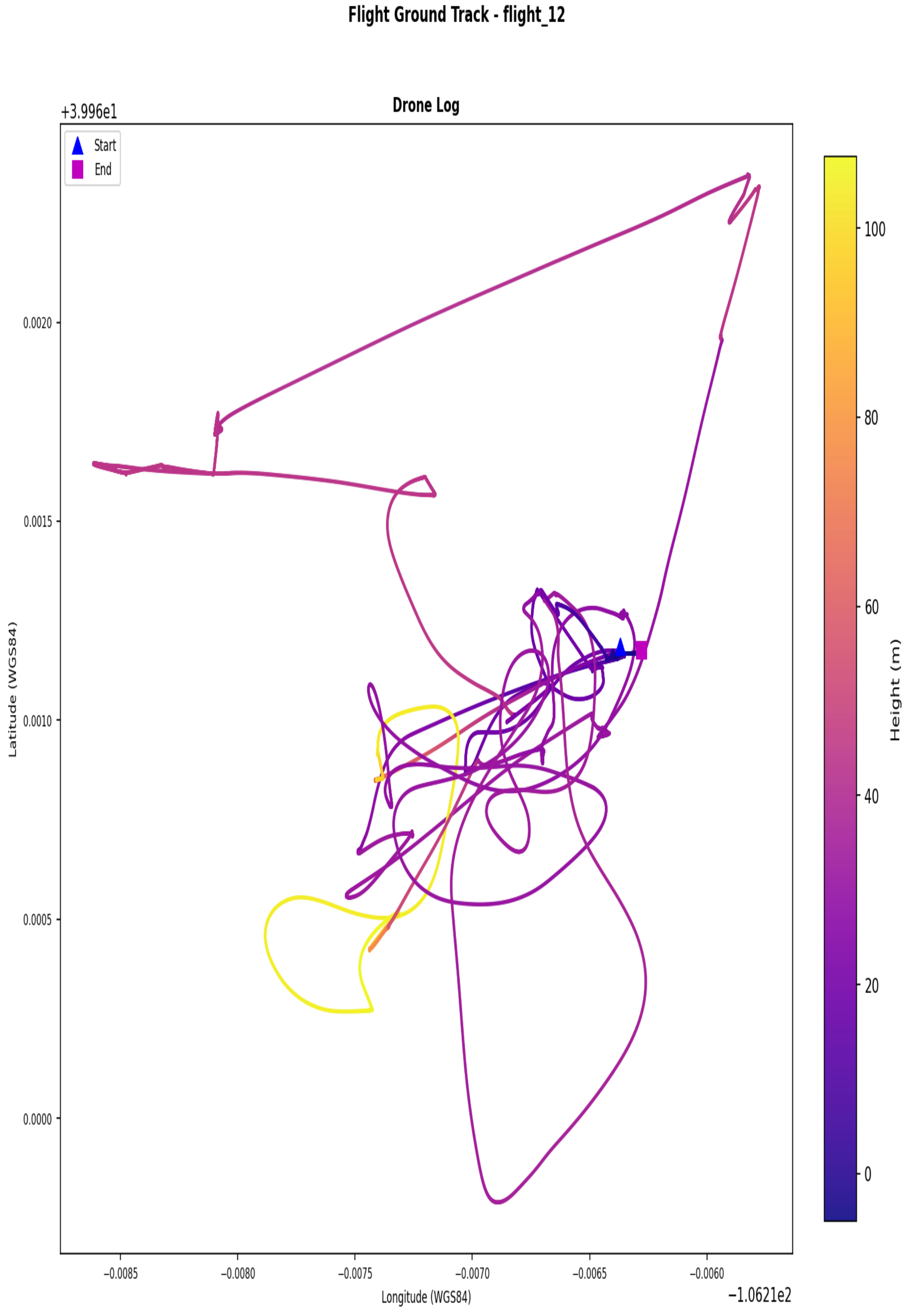
Timestamp (UTC)	Source	Event	Details
2018-06-21 15:51:52	drone_flight_storage:dl	Log started	P3S
2018-06-21 16:02:22	drone_flight_storage:dl	Reached peak height	107.5 m
2018-06-21 16:07:30	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY017.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).
A KML file (FLY017.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 13

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `483cbf7d95319254e6ec429a85882b9743bf93d202af4a08fa8d53691041b530`
Drone Log Name: FLY018.csv
Start: 2018-06-21T19:51:18.850000+00:00
End: 2018-06-21T19:52:57.875000+00:00
Duration: 99.0 s

Flight Metrics

Peak Height: 1.3970570316977093 m
Distance Travelled: 11381919.776069224 m
Video Recording: No
Photo Capture: No

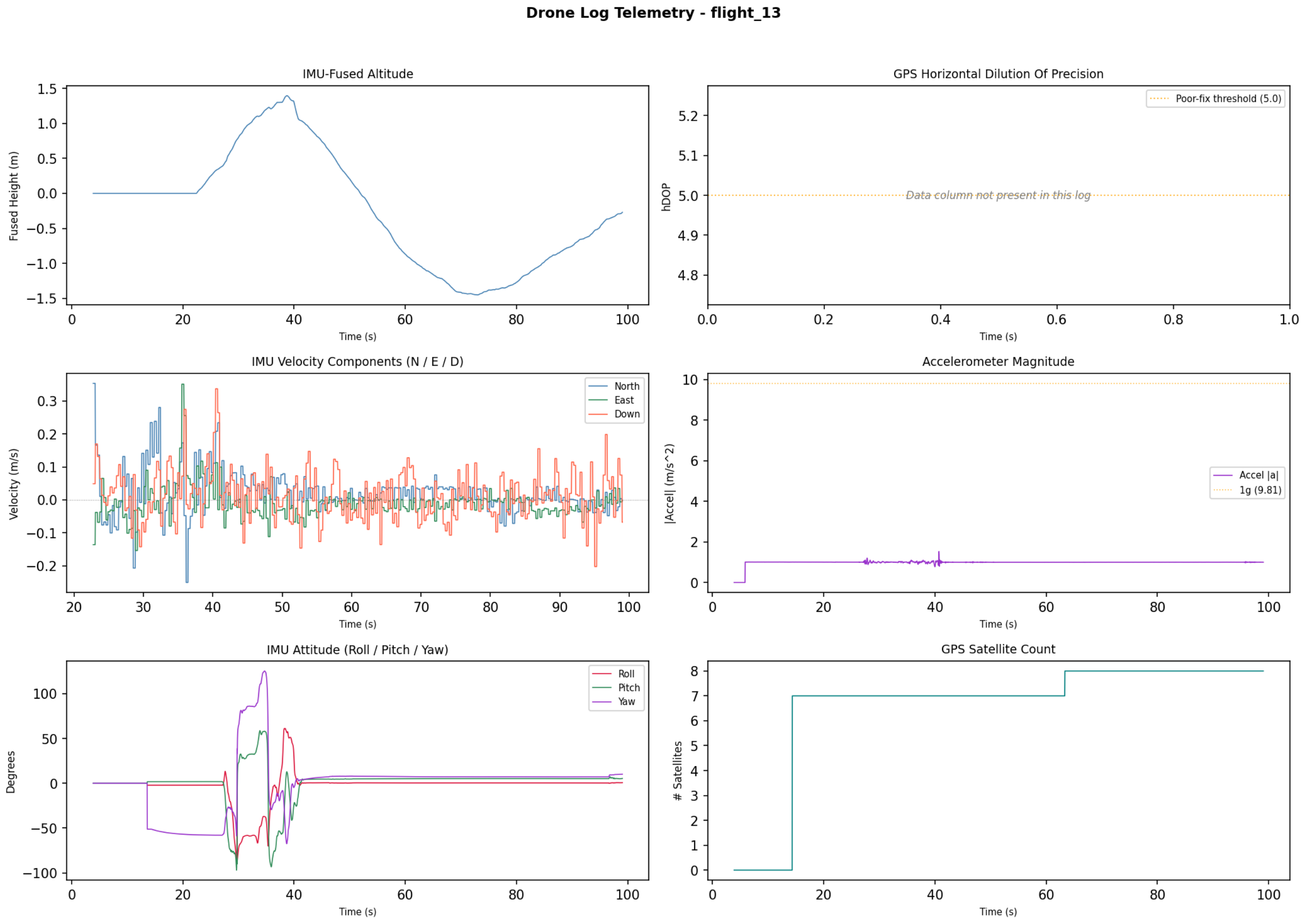
Fly Modes Observed (Chronological)

1. 0

Key Events

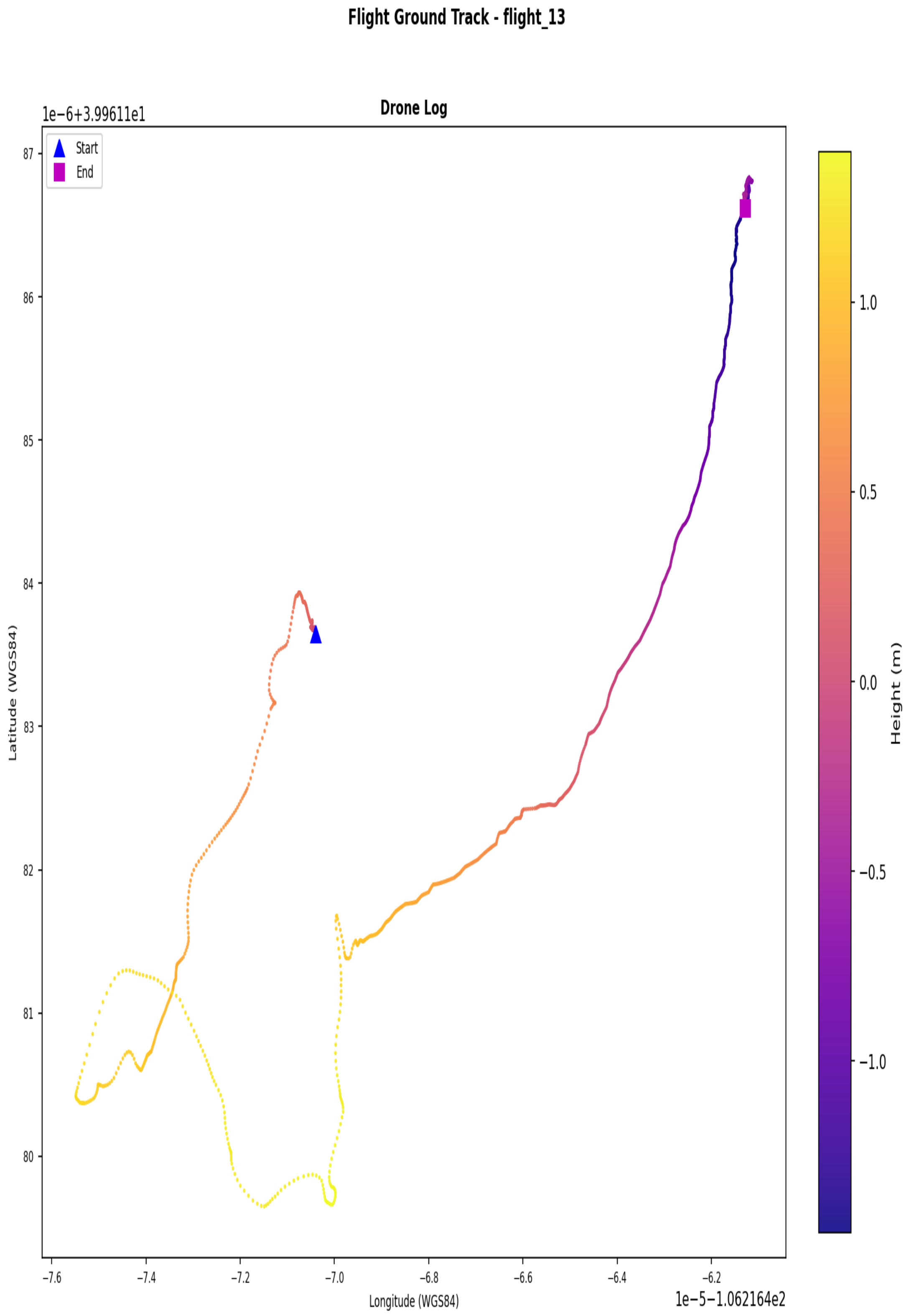
Timestamp (UTC)	Source	Event	Details
2018-06-21 19:51:47	drone_flight_storage:dl	Log started	P3S
2018-06-21 19:51:57	drone_flight_storage:dl	Reached peak height	1.4 m
2018-06-21 19:52:57	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY018.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY018.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 14

Correlation and Flight Window

Status:	Unmatched (no drone log corroboration)
Drone Log SHA-256:	`9fbb6503b83f372741133cf21e5bd8c0d494755ccde59cf8fb721b334a5e85c5`
Drone Log Name:	FLY019.csv
Start:	2018-06-21T19:56:18.117000+00:00
End:	2018-06-21T20:12:31.455000+00:00
Duration:	973.3 s

Flight Metrics

Peak Height:	8.298943365338397 m
Distance Travelled:	195.12062621097402 m
Video Recording:	No
Photo Capture:	No

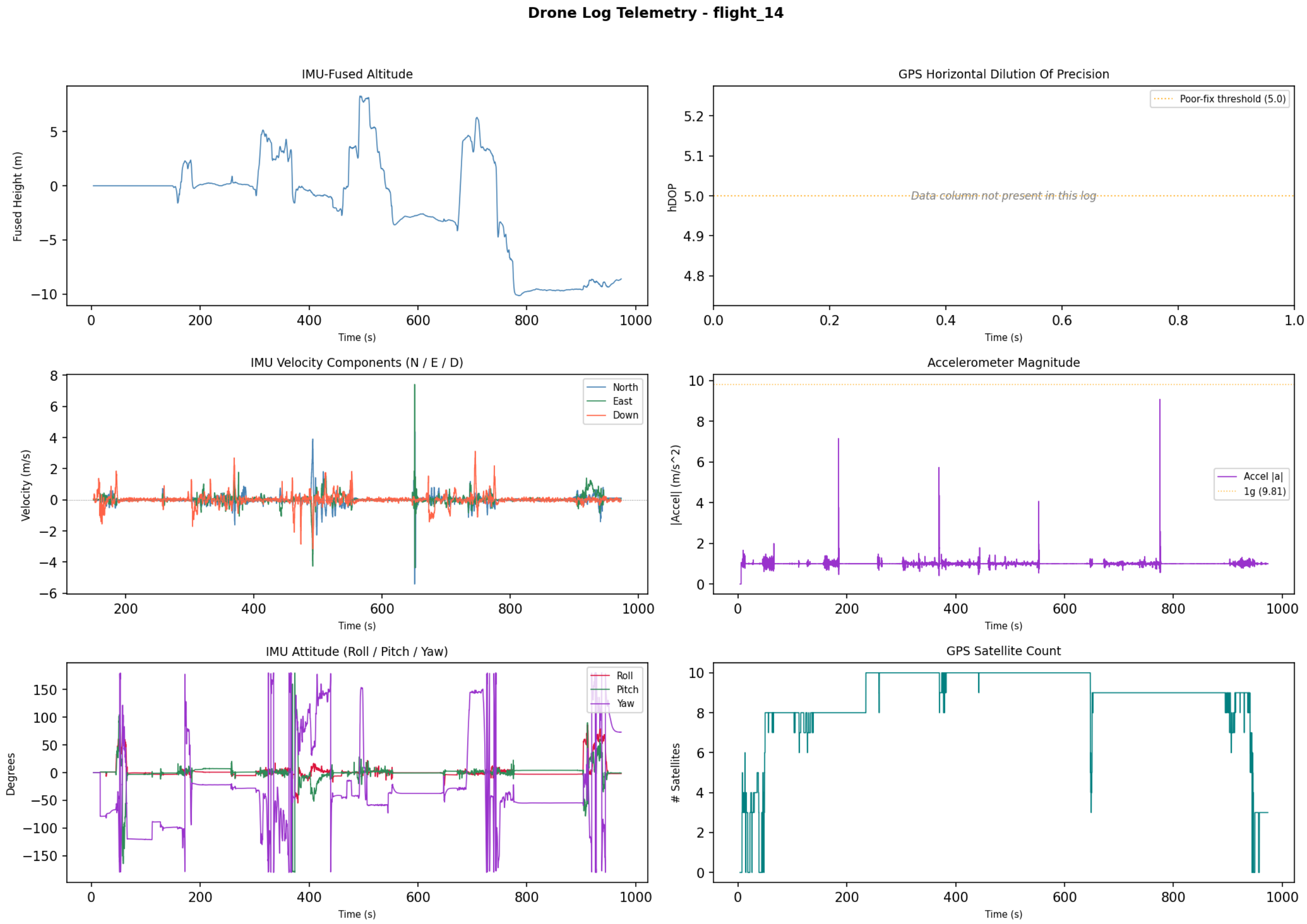
Fly Modes Observed (Chronological)

1. 0
2. 1
3. 2
4. 3

Key Events

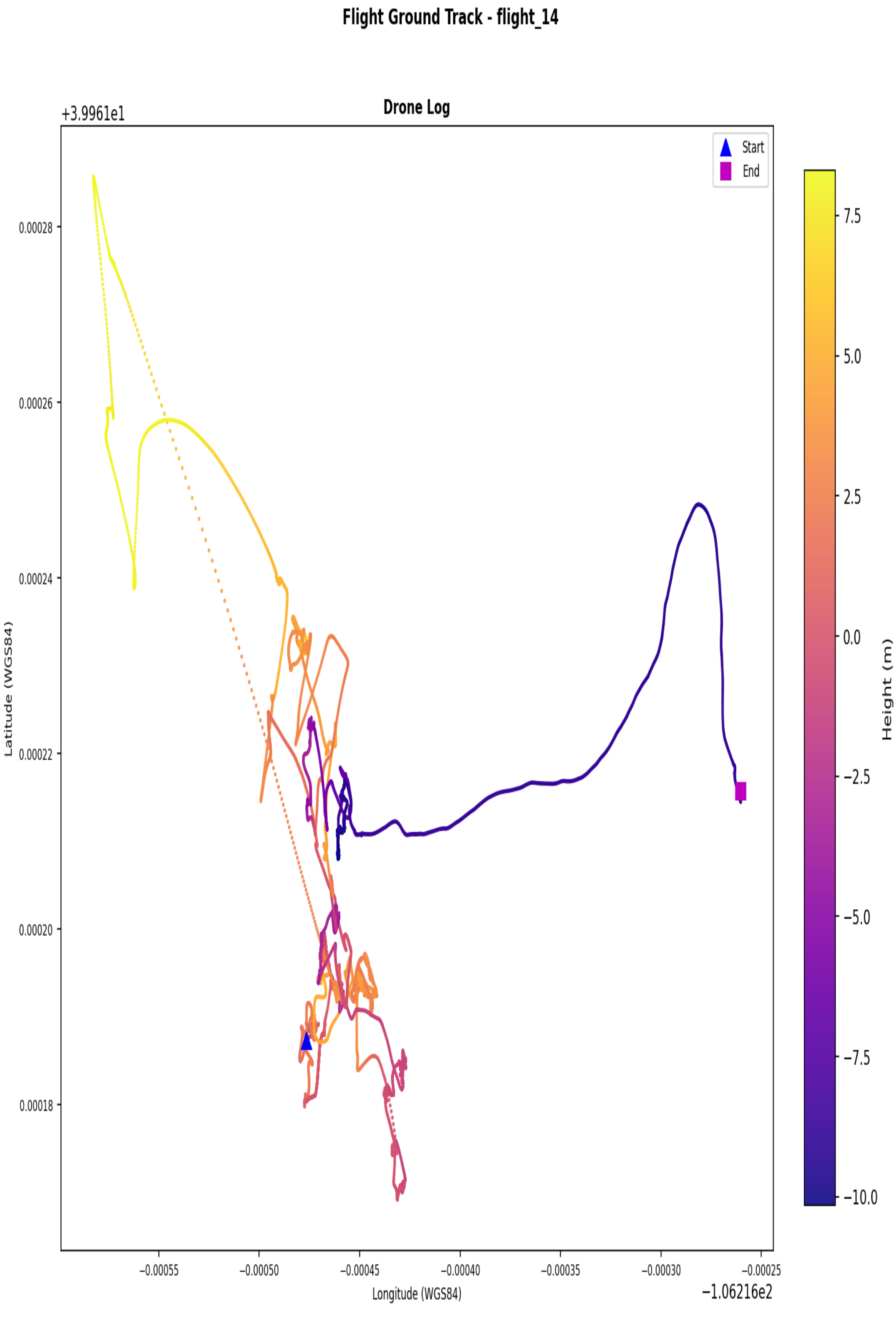
Timestamp (UTC)	Source	Event	Details
2018-06-21 19:56:22	drone_flight_storage:dl	Log started	P3S
2018-06-21 20:04:31	drone_flight_storage:dl	Reached peak height	8.3 m
2018-06-21 20:12:31	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY019.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY019.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 15

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `ce628b7ec2f92ed76df22d3c8b6fc73623375032b341358ef2a6adb619485767`
Drone Log Name: FLY020.csv
Start: 2018-06-21T20:12:38.107000+00:00
End: 2018-06-21T20:14:39.845000+00:00
Duration: 121.7 s

Flight Metrics

Peak Height: -
Distance Travelled: 7.562245330351284 m
Video Recording: No
Photo Capture: No

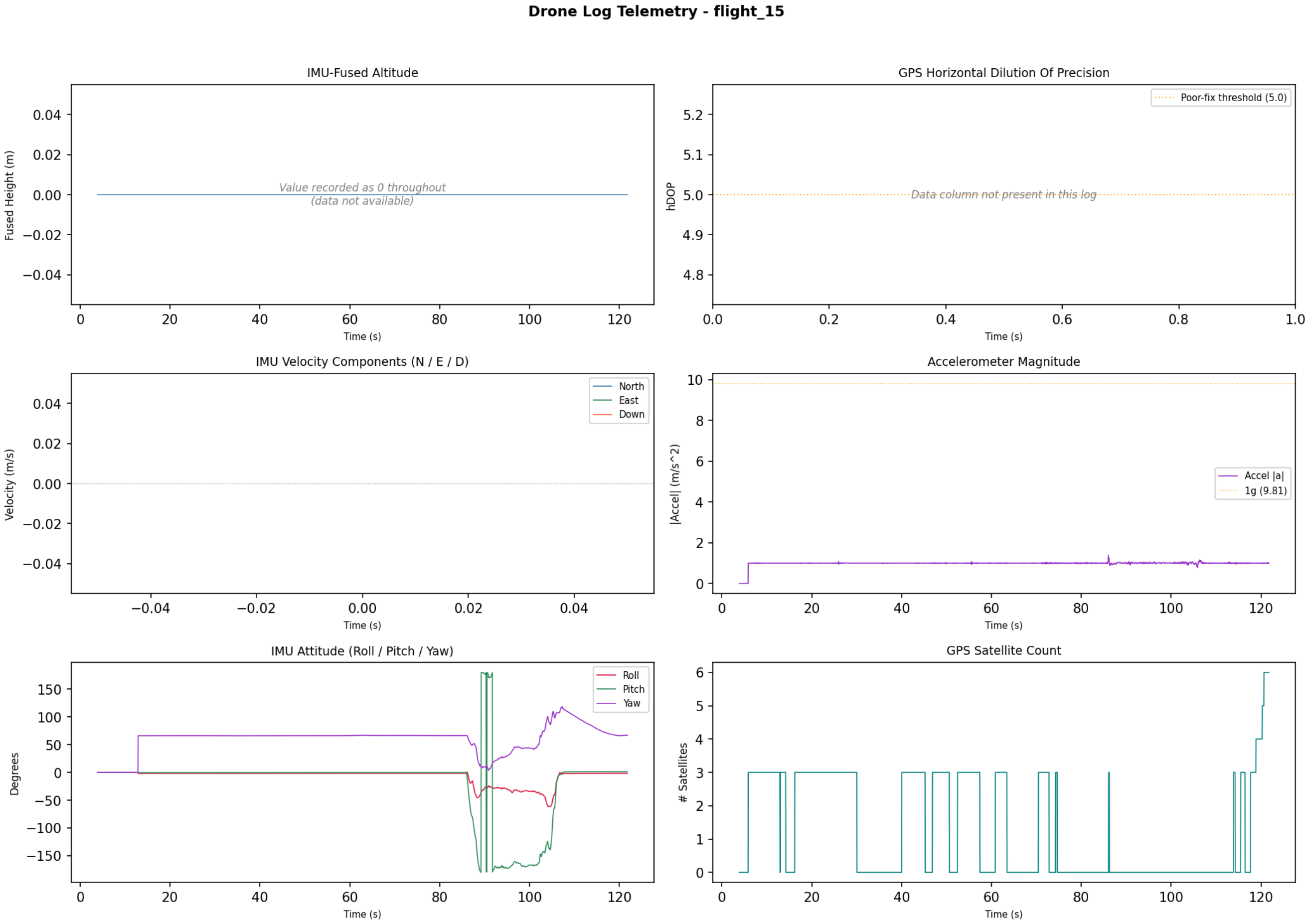
Fly Modes Observed (Chronological)

1. 0

Key Events

Timestamp (UTC)	Source	Event	Details
2018-06-21 20:12:42	drone_flight_storage:dl	Log started	P3S
2018-06-21 20:12:42	drone_flight_storage:dl	Reached peak height	-0.0 m
2018-06-21 20:14:43	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 16

Correlation and Flight Window

Status:	Unmatched (no drone log corroboration)
Drone Log SHA-256:	`fac65daad99468629cc23b193aea2619860be905c8db00074f43536f950ef4d3`
Drone Log Name:	FLY021.csv
Start:	2018-06-21T20:14:47.362000+00:00
End:	2018-06-21T20:15:40.185000+00:00
Duration:	52.8 s

Flight Metrics

Peak Height:	-
Distance Travelled:	-
Video Recording:	No
Photo Capture:	No

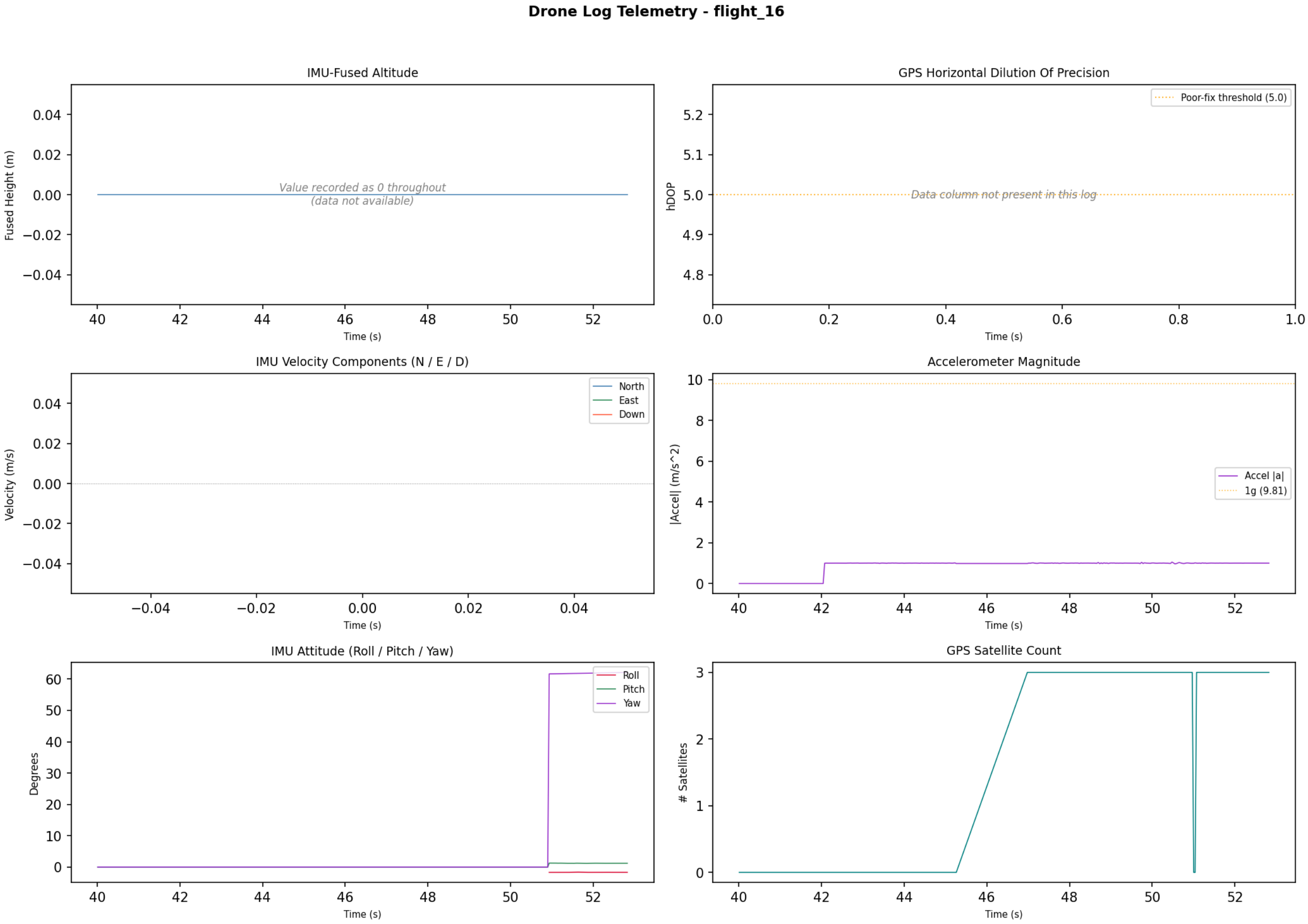
Fly Modes Observed (Chronological)

1. 0

Key Events

Timestamp (UTC)	Source	Event	Details
2018-06-21 20:14:48	drone_flight_storage:dl	Log started	P3S
2018-06-21 20:14:49	drone_flight_storage:dl	Reached peak height	-0.0 m
2018-06-21 20:15:40	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	No
controller_ios	Yes
drone_sd	No
drone_flight_storage	Yes

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
controller_ios	2	14	0	6	1	0	0
drone_flight_storage	0	0	22	0	0	0	0

Data Quality Notes per Category

Databases (14 files):

- 8 database(s) empty (controller_ios)

Drone Logs (22 files):

- 22 file(s) with row anomalies
- 22 file(s) with empty columns
- 22 file(s) with constant-value columns

Flight Logs (6 files):

- formats: 3 bracketed_log, 3 json_array_log

Flight Records (1 files):

- 1 file(s) with row anomalies
- 1 file(s) with empty columns
- 1 file(s) with constant-value columns

6. Data Quality and Anomalies

Data Quality Observations

- Drone_logs: 22 file(s) with row anomalies detected in telemetry data.
- Drone_logs: 22 file(s) with empty columns - some telemetry channels contain no data.
- Drone_logs: 22 file(s) with constant-value columns - some telemetry channels may be unreliable.
- Flight_logs: log files present in the following format(s): 3 bracketed_log, 3 json_array_log.
- Flight_records: 1 file(s) with row anomalies detected in telemetry data.
- Flight_records: 1 file(s) with empty columns - some telemetry channels contain no data.
- Flight_records: 1 file(s) with constant-value columns - some telemetry channels may be unreliable.

Pipeline Anomaly Flags

1. [p2 - controller android]: source evidence not found
2. [p3 - controller android]: source evidence not found
3. [p3 - controller ios - drone logs]: no artefacts found
4. [p3 - controller ios - images]: no artefacts found
5. [p3 - controller ios - videos]: no artefacts found
6. [p3 - drone sd]: source evidence not found
7. [p4 - controller android]: source evidence not found
8. [p4 - drone sd]: source evidence not found
9. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY000.csv
10. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY001.csv
11. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY002.csv
12. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY003.csv
13. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY004.csv
14. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY014.csv

Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Databases

5255DF85724869CD36D4AFCD12AE092D.db

- database_empty: all tables empty

djiFMDB.db

- database_empty: all tables empty

djiFMDB_1.db

- database_empty: all tables empty

dji_fs_upgrade_local_cache.db

- database_empty: all tables empty

flysafe_app_dynamic_areas.db

- database_empty: all tables empty

flysafe_app_dynamic_areas_1.db

- database_empty: all tables empty

flysafe_dji_flight_dynamic_areas.db

- database_empty: all tables empty

flysafe_dji_flight_dynamic_areas_1.db

- database_empty: all tables empty

Drone Logs

norm_FLY000.csv

- missing_gps: 47197 row(s) affected
- contains_no_value: 27 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+22 more)

- contains_constant_value: 62 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+57 more)
- quaternion_invalid: 86 row(s) affected

norm_FLY001.csv

- missing_gps: 8270 row(s) affected
- contains_no_value: 27 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+22 more)
- contains_constant_value: 63 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+58 more)

norm_FLY002.csv

- missing_gps: 8300 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+21 more)
- contains_constant_value: 62 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+57 more)
- quaternion_invalid: 257 row(s) affected

norm_FLY003.csv

- missing_gps: 679 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+21 more)
- contains_constant_value: 71 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+66 more)
- quaternion_invalid: 255 row(s) affected

norm_FLY004.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 3821 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+21 more)
- contains_constant_value: 65 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+60 more)
- quaternion_invalid: 228 row(s) affected

norm_FLY005.csv

- missing_gps: 7559 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+21 more)
- contains_constant_value: 57 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+52 more)
- quaternion_invalid: 838 row(s) affected

norm_FLY006.csv

- missing_gps: 3919 row(s) affected
- contains_no_value: 23 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+18 more)
- contains_constant_value: 48 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+43 more)
- quaternion_invalid: 257 row(s) affected

norm_FLY007.csv

- missing_gps: 3730 row(s) affected
- contains_no_value: 23 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+18 more)
- contains_constant_value: 48 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+43 more)
- quaternion_invalid: 257 row(s) affected

norm_FLY008.csv

- missing_gps: 41927 row(s) affected
- altitude_negative: 41 row(s) affected

- contains_no_value: 10 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+5 more)
- contains_constant_value: 14 column(s) - AirCraftCondition, AirCraftCondition:launch_state, Battery(0):maxVolts, GPS(0):numGLNAS, GPS(0):numGPS (+9 more)
- quaternion_invalid: 435 row(s) affected
- attitude_out_of_bounds: 3135 row(s) affected

norm_FLY009.csv

- missing_gps: 10166 row(s) affected
- contains_no_value: 12 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+7 more)
- contains_constant_value: 33 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+28 more)
- quaternion_invalid: 257 row(s) affected
- attitude_out_of_bounds: 90 row(s) affected

norm_FLY010.csv

- missing_gps: 245 row(s) affected
- contains_no_value: 34 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+29 more)
- contains_constant_value: 65 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+60 more)
- quaternion_invalid: 204 row(s) affected

norm_FLY011.csv

- missing_gps: 2515 row(s) affected
- contains_no_value: 12 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+7 more)
- contains_constant_value: 37 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+32 more)
- quaternion_invalid: 477 row(s) affected
- attitude_out_of_bounds: 13 row(s) affected

norm_FLY012.csv

- missing_gps: 3386 row(s) affected
- contains_no_value: 12 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+7 more)
- contains_constant_value: 32 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+27 more)
- quaternion_invalid: 257 row(s) affected

norm_FLY013.csv

- missing_gps: 8361 row(s) affected
- altitude_negative: 1247 row(s) affected
- contains_no_value: 10 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+5 more)
- contains_constant_value: 17 column(s) - AirCraftCondition, Battery(0):maxVolts, Battery(0):minCurrent, Battery(0):minWatts, GPS(0):Date (+12 more)
- quaternion_invalid: 257 row(s) affected

norm_FLY014.csv

- missing_gps: 297 row(s) affected
- contains_no_value: 32 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+27 more)
- contains_constant_value: 87 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+82 more)
- quaternion_invalid: 256 row(s) affected

norm_FLY015.csv

- missing_gps: 15807 row(s) affected
- altitude_negative: 1449 row(s) affected
- contains_no_value: 10 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C,

General:relativeHeight:C (+5 more)
- contains_constant_value: 17 column(s) - AirCraftCondition, AirCraftCondition:launch_state, Battery(0):maxVolts, Battery(0):minCurrent, Battery(0):minWatts (+12 more)
- quaternion_invalid: 300 row(s) affected
- attitude_out_of_bounds: 52 row(s) affected

norm_FLY016.csv
- missing_gps: 3176 row(s) affected
- altitude_negative: 1401 row(s) affected
- contains_no_value: 10 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+5 more)
- contains_constant_value: 41 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+36 more)
- quaternion_invalid: 262 row(s) affected
- attitude_out_of_bounds: 79 row(s) affected

norm_FLY017.csv
- missing_gps: 27096 row(s) affected
- altitude_negative: 2172 row(s) affected
- contains_no_value: 9 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+4 more)
- contains_constant_value: 16 column(s) - AirCraftCondition, Battery(0):maxVolts, Battery(0):minCurrent, Battery(0):minWatts, GPS(0):numGLNAS (+11 more)
- quaternion_invalid: 170 row(s) affected

norm_FLY018.csv
- missing_gps: 2760 row(s) affected
- altitude_negative: 1315 row(s) affected
- contains_no_value: 10 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+5 more)
- contains_constant_value: 41 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+36 more)
- quaternion_invalid: 278 row(s) affected
- attitude_out_of_bounds: 10 row(s) affected

norm_FLY019.csv
- missing_gps: 27618 row(s) affected
- altitude_negative: 13890 row(s) affected
- contains_no_value: 10 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+5 more)
- contains_constant_value: 16 column(s) - AirCraftCondition, Battery(0):minCurrent, Battery(0):minWatts, GPS(0):Date, GPS(0):numGLNAS (+11 more)
- quaternion_invalid: 357 row(s) affected
- attitude_out_of_bounds: 418 row(s) affected

norm_FLY020.csv
- missing_gps: 3361 row(s) affected
- contains_no_value: 24 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+19 more)
- contains_constant_value: 41 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+36 more)
- quaternion_invalid: 257 row(s) affected
- attitude_out_of_bounds: 505 row(s) affected

norm_FLY021.csv
- missing_gps: 365 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+21 more)
- contains_constant_value: 59 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+54 more)
- quaternion_invalid: 264 row(s) affected

Flight Logs

2018-06-19_13_16_58-03Z1389682.dat

- format: json_array_log | entries: 21

2018-06-19_13_18_23-03Z1389682.dat

- format: json_array_log | entries: 1

2018-06-19_13_18_49-03Z1389682.dat

- format: json_array_log | entries: 2

upgradeLog_0.txt

- format: bracketed_log | entries: 6

upgradeLog_0_1.txt

- format: bracketed_log | entries: 24

upgradeLog_1.txt

- format: bracketed_log | entries: 6

Flight Records

norm_DJIFlightRecord_2018-06-19__13-16-58_.csv

- altitude_negative: 89 row(s) affected
- contains_no_value: 29 column(s) - APP_SER_WARN.warn, CENTER_BATTERY.isBatteryOnCharge, CENTER_BATTERY.voltageCell5 [V], CUSTOM.isPhoto, DEFORM.deformMode (+24 more)
- contains_constant_value: 173 column(s) - CAMERA_INFO.cameraType, CAMERA_INFO.connectState, CAMERA_INFO.enabledPhoto, CAMERA_INFO.encryptStatus, CAMERA_INFO.firmErrorType (+168 more)

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Drone Model inconsistency detected - conflicting values observed: P3 Standard, P3C, P3S. Manual verification required.
2. DJI Application Version inconsistency detected - conflicting values observed: 3.1.38, 4.2.20. Manual verification required.
3. Product Type inconsistency detected - conflicting values observed: 2, 7. Manual verification required.
4. Note: no firmware version found - Device and account information.
5. Note: no last known latitude (find aircraft) found - Device and account information.
6. Note: no last known longitude (find aircraft) found - Device and account information.
7. Note: no last known altitude (find aircraft) found - Device and account information.
8. Note: no last known heading (find aircraft) found - Device and account information.
9. Note: no last known horiz. accuracy (find aircraft) found - Device and account information.
10. 15 flight record(s) could not be correlated to a drone log (flight_01, flight_02, flight_03, flight_04, flight_06, flight_07, flight_08, flight_09, flight_10, flight_11, flight_12, flight_13, flight_14, flight_15, flight_16). Independent corroboration is absent for these flight(s).
11. 6 database file(s) contain records and should be examined for relevant artefacts.

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

com.dji.go.plist		Account Data	controller_ios	extract_logical
SHA-256:	0a9db04d880caabcc772b10c810e9653093be0441a341e05276993af164d3478			
SHA-1:	10b0a43ff6d4dae62490f4d7f0f0fb31f743c87e			

com.dji.pilot.plist		Account Data	controller_ios	extract_logical
SHA-256:	22250fc6302bcf76a30c3af7a7ff06b7edea395d34156631db6299403058cc87			
SHA-1:	4c4e1eb46077443bebe4a8c0d98a3ac2767479ad			

5255DF85724869CD36D4AFCD12AE092D.db		Databases	controller_ios	extract_logical
SHA-256:	aae50532bee18355508de941e178ea847cd725018bb03a93589944f7cad4898e			
SHA-1:	7ecb7ed8a4b0ecf3028259a0eb0dcee998fcbc0c			

538103E3AB6E8F987EC10913E64A3E24.db		Databases	controller_ios	extract_logical
SHA-256:	6df74165eb44a8570bdc89f0038e0831de6ba509aa2aa2bd15aa1a6a2786cea0			
SHA-1:	f2d9e8167faa28bbb2685b330a7fabe2732943aa			

djiFMDB.db		Databases	controller_ios	extract_logical
SHA-256:	5fdae73b94d5e7a078a8bfd1afda762eae9a5bff869139e9242dfecc749fd52a			
SHA-1:	b1d200f2a2416e3c776ebc4e1779312198adf52b			

dji_fs_upgrade_local_cache.db		Databases	controller_ios	extract_logical
SHA-256:	160afa85388b924b8dcbb457db44c630a72905aff9e3d6ebcdf2eab6ded1ab49			
SHA-1:	c9a66614704ea8d5b877afc7ddad8f99b4a2d056			

flysafe_app_dynamic_areas.db		Databases	controller_ios	extract_logical
SHA-256:	483da04ac8232310746927e831150673403cb79e12dbedf322869a1317fa7fcf			
SHA-1:	4e8e284810efa83aa0861217939fff02c9d7ad12			

flysafe_areas_djigo_en.db		Databases	controller_ios	extract_logical
SHA-256:	1a1c024072fba37bffbf1198f4227fcf2f4ffad347ad9a93b164a7077cd0b6f8			
SHA-1:	761bad6393f8f8a28ecea57069626b9a9eaa6de4			

flysafe_dji_flight_dynamic_areas.db		Databases	controller_ios	extract_logical
SHA-256:	6d434e165f689c505c6d051cc3ba0aa7dd26737023ff1aee562fd88f106a8953			
SHA-1:	80d3466fd6360f9edab821a05cfbf97ac0ce6804			

flysafe_license_unlock.db		Databases	controller_ios	extract_logical
SHA-256:	b78b51c568903848432755f60acb4c1a6225953b18e4a533f4fb5360c70d65a6			
SHA-1:	e128ff5c503caf2a906e9f80ab86feb342d7237f			

flysafe_static_circle.db		Databases	controller_ios	extract_logical
SHA-256:	4787fc859c865dab6b3d0db8eb68a55c764ca5d13105b14429cc2d455708cfb4			
SHA-1:	ccf2d5995e798abe1f1255c252c28ec8ae699dd4			

djiFMDB_1.db		Databases	controller_ios	extract_logical
SHA-256:	5fdae73b94d5e7a078a8bfd1afda762eae9a5bff869139e9242dfecc749fd52a			
SHA-1:	b1d200f2a2416e3c776ebc4e1779312198adf52b			

flysafe_app_dynamic_areas_1.db	Databases	controller_ios	extract_logical
SHA-256:	ce02ae43e9061d526429d75f4aab106e9dc0133a90ac005df05e984ccf52e6fe		
SHA-1:	5c25a94e52815ddf092eccb33456a159cc66f577		

flysafe_areas_djigo.db	Databases	controller_ios	extract_logical
SHA-256:	b5a6d15e6e04e9cbcc0902a392540a2b25d2a9722c6219d5cf34dfad90f838c8		
SHA-1:	a2836e85087f06b331fbfea45025b05cf35f5942		

flysafe_dji_flight_dynamic_areas_1.db	Databases	controller_ios	extract_logical
SHA-256:	56ef148210fceffbca82dc86b201fbdc340f32c9a247b9341ba85c54a21918c6		
SHA-1:	91649bae84edd1c230392932c2a5ad0e917b0f7c		

flysafe_static_circle_1.db	Databases	controller_ios	extract_logical
SHA-256:	4787fc859c865dab6b3d0db8eb68a55c764ca5d13105b14429cc2d455708cfb4		
SHA-1:	ccf2d5995e798abe1f1255c252c28ec8ae699dd4		

upgradeLog_0.txt	Flight Logs	controller_ios	extract_logical
SHA-256:	aadaf277c7c030c449166789562b4d39f33f67162c9f058fe400349b58d56b66		
SHA-1:	342d87657eb15a9c7c784120744419d0bbafff1d		

2018-06-19_13_16_58-03Z1389682.dat	Flight Logs	controller_ios	extract_logical
SHA-256:	790772e2dc601d7f721b92bc91f5ce6f1c3e804ccc20dbe5ce10db81a2e77023		
SHA-1:	9fa9371ade21fc045c6916a8eb417ffd01b07b8b		

2018-06-19_13_18_23-03Z1389682.dat	Flight Logs	controller_ios	extract_logical
SHA-256:	3b54b251405cd0ce4832ee6e7efb646837e6f9748c2167e8b0986a9582ef1ebd		
SHA-1:	24e32b4d1bcacb5a96e663e5b972d0f53de1ddf9		

2018-06-19_13_18_49-03Z1389682.dat	Flight Logs	controller_ios	extract_logical
SHA-256:	025c304a3901d33f4a705b5bd8ee765907a1c03dbfc8e7aa74d7882c53519d82		
SHA-1:	79aefb32b9f58b5db1525f352b535b57c0e426e3		

upgradeLog_0_1.txt	Flight Logs	controller_ios	extract_logical
SHA-256:	d5cbf55de331a3c55515c49146f50db21389a693289857be3e98e65a2ca240ff		
SHA-1:	f2a0453de58760a3969f57242a9d1c5444d7a5c3		

upgradeLog_1.txt	Flight Logs	controller_ios	extract_logical
SHA-256:	975f26503bba8e3b4a41fdc144514bcd55e3dfaadb5b597c673c6b6e7105daf3		
SHA-1:	8403babb7d66204207b47d5e7752f9bb80301bdd		

DJIFlightRecord_2018-06-19__13-16-58_.txt	Flight Records	controller_ios	extract_logical
SHA-256:	f38be7c355fd16890c06e28602795aac66b425fa624e4edae05c76083ee67268		
SHA-1:	bb11174deec0b71cc7a7e862947ed6331500892c		

FLY000.DAT	Drone Logs	drone_flight_storage	extract_logical
SHA-256:	4ef04116a2601f795d11d13e37924f7809d77f77255178c9fc4d4ead74073e81		
SHA-1:	03a1448280db64f7228dfeda55cfd1264bdd18db		

FLY001.DAT	Drone Logs	drone_flight_storage	extract_logical
SHA-256:	29734f5dbafcbd7675a3d299f27b382045fcc3bef95e3afadc3f74da7dfe60dc		
SHA-1:	a3abce32ab37fb6a2bdcecd4a0c26268f0bb6e02		

FLY002.DAT		Drone Logs	drone_flight_storage	extract_logical
SHA-256:	0de262d6ef03056c48e5ae387a46faa6ff0c62e3080738ec49373f47d39e83d8			
SHA-1:	4d4627b4211dc07ebce661c73c46d87d534e9857			

FLY003.DAT		Drone Logs	drone_flight_storage	extract_logical
SHA-256:	d41201c54dd0309c910c0f1d0dc8520bbdd65bc2589a3a521ab3c534b039fcbe			
SHA-1:	75e39760401d2d1208be3b30554e1d9ecc627f08			

FLY004.DAT		Drone Logs	drone_flight_storage	extract_logical
SHA-256:	87431f5861ce4cec730239a79332c416d98692ecda932c6d8be86f12888678d2			
SHA-1:	c3d16f9f1ba262feb4c20252fc6d7d1b7510cea2			

FLY005.DAT		Drone Logs	drone_flight_storage	extract_logical
SHA-256:	e9db822cb9c76eb64676283c6b60640065b923e345dec124714c222f5f93609f			
SHA-1:	54c4d5bfa7d1cd571ea143acd821bde41b71582a			

FLY006.DAT		Drone Logs	drone_flight_storage	extract_logical
SHA-256:	9ea14e35558b4e433c255e3e6bfd8dcf9c7f483e924f377194c3a5d984ecf384			
SHA-1:	dde1b2ba0cb1666ddc1af6b8d58b799516e7a681			

FLY007.DAT		Drone Logs	drone_flight_storage	extract_logical
SHA-256:	4c770abc022a1797f7d1eef4182fa2bead4bb6663f5f1ef0e0b2a9903666a0f8			
SHA-1:	823dbe0b5bad6417616180f8cbe26d778f70ed42			

FLY008.DAT		Drone Logs	drone_flight_storage	extract_logical
SHA-256:	7831a998afb0ee5203ec560962c0522a91e06087b005640846c72265e9ece228			
SHA-1:	fa7894eefcffb28af65d0f25461e3e6ae76197a4			

FLY009.DAT		Drone Logs	drone_flight_storage	extract_logical
SHA-256:	33857c854395cf6ef00e5511a7d361bf8d30942a47040d1194e09fe466aafc2e			
SHA-1:	059475c3e4ffd60ca65004b7647dc5841d865973			

FLY010.DAT		Drone Logs	drone_flight_storage	extract_logical
SHA-256:	b398ea030b651202f96d68179de4090a55499734d974acd61ec331fc1cacab43			
SHA-1:	b2092f0dd395fbfa028d0c64043514ea1d70361c			

FLY011.DAT		Drone Logs	drone_flight_storage	extract_logical
SHA-256:	67e78ae79cdb21d4fae9c4a42eb00c89852b827790d134dde4c1c49b11c921b2			
SHA-1:	17e95b242784290c5481c6bf41ec06175d3eeed6			

FLY012.DAT		Drone Logs	drone_flight_storage	extract_logical
SHA-256:	0056b3201a3675a1f2bdf6828458d0b7d6ce40f6176c70713b5e2129c77e1f17			
SHA-1:	ad42dd7196d12f374b0e4f5bcf30b9453a933df1			

FLY013.DAT		Drone Logs	drone_flight_storage	extract_logical
SHA-256:	ca5d32c4fcb6a727bd992e3cc10a3a75dcd2b45aa6e80d9c1c12a54894bfed0b			
SHA-1:	e3a187a48a3aee3e819dc295e043153579cefed9			

FLY014.DAT		Drone Logs	drone_flight_storage	extract_logical
SHA-256:	56cdd28dbb2cd7949778d26aea0e9f62e49ebe51752ddaac6609af84b3d5f216			
SHA-1:	23594187b7df1d08e7964ba42c2071da62d5bd57			

FLY015.DAT		**Drone Logs**	**drone_flight_storage**	**extract_logical**
SHA-256:	ec937557175b7c83d6e57ced8fad1a8241210e5c1aefc34a646165d9f7c12fbc			
SHA-1:	1d33bcd18e4ef8ea010f4f6893f327c3b235550d			

FLY016.DAT		**Drone Logs**	**drone_flight_storage**	**extract_logical**
SHA-256:	b29e39bcfeb94274938e4670c541763b2450e5ef58ffd242056e4266e2d5d46e			
SHA-1:	45749aa63727ba77b38d6efb79dc8712a4b99ed7			

FLY017.DAT		**Drone Logs**	**drone_flight_storage**	**extract_logical**
SHA-256:	858ba001e4c0b0fb0328ffbaa1a6f6bab3427244014d6f3769a742e3e96739cc			
SHA-1:	65403458212449c56983d896f0ea8df0042cce05			

FLY018.DAT		**Drone Logs**	**drone_flight_storage**	**extract_logical**
SHA-256:	0854ca8d4571be8af0397baf76382d030f34b696129f0691f794ed8be7b5a8a7			
SHA-1:	62d36bdef3c715248bb657d0732f8733cceb15b6			

FLY019.DAT		**Drone Logs**	**drone_flight_storage**	**extract_logical**
SHA-256:	f7445b78f56dd84f6d349b3b0ec7c0e32a198bf88cc796e5f6a9f145bf00940f			
SHA-1:	2643fa811601104309aacee1300036409cb0e784			

FLY020.DAT		**Drone Logs**	**drone_flight_storage**	**extract_logical**
SHA-256:	96cf3185397ebd447880f6e184fc6429561032d6d60eb8e7fe821dcc11a0983d			
SHA-1:	ebf6d1046c7f4c4c6beeeff4470756bb2d011f44			

FLY021.DAT		**Drone Logs**	**drone_flight_storage**	**extract_logical**
SHA-256:	2bd4275ffb9ee3138fb2c1f94ac14694f32fb1821b4a19b251f8c57776e3396f			
SHA-1:	0aefd07c1306aacdd0557582945874802424ec13			

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Controller Ios

Category	P3 Extracted	P4 Decoded	P5 Normalised
Account Data	2	2	0
Databases	14	0	0
Flight Logs	6	0	0
Flight Records	1	1	1

Drone Flight Storage

Category	P3 Extracted	P4 Decoded	P5 Normalised
Drone Logs	22	44	22

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-15 15:16:18 parser_ios v- rc=n/a

cmd: ios_logical.zip C:\Users\Floris\Documents\dfdof_output\CASE-df001-2018-ios\p2_image_parsing\controller_ios_parsed

out: controller_ios_parsed

2026-06-15 15:16:39 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2018-06-19__13-16-58_.txt DJIFlightRecord_2018-06-19__13-16-58_.csv

out: DJIFlightRecord_2018-06-19__13-16-58_.csv

2026-06-15 15:21:13 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY000.DAT

out: FLY000.csv, FLY000.kml

2026-06-15 15:21:13 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY001.DAT

out: FLY001.csv, FLY001.kml

2026-06-15 15:21:14 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY002.DAT

out: FLY002.csv, FLY002.kml

2026-06-15 15:21:15 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY003.DAT

out: FLY003.csv, FLY003.kml

2026-06-15 15:21:15 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY004.DAT

out: FLY004.csv, FLY004.kml

2026-06-15 15:21:16 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY005.DAT

out: FLY005.csv, FLY005.kml

2026-06-15 15:21:17 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY006.DAT

out: FLY006.csv, FLY006.kml

2026-06-15 15:21:18 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY007.DAT

out: FLY007.csv, FLY007.kml

2026-06-15 15:21:18 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY008.DAT

out: FLY008.csv, FLY008.kml

2026-06-15 15:21:19 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY009.DAT

out: FLY009.csv, FLY009.kml

2026-06-15 15:21:20 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY010.DAT

out: FLY010.csv, FLY010.kml

2026-06-15 15:21:21 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY011.DAT

out: FLY011.csv, FLY011.kml

2026-06-15 15:21:22 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY012.DAT

out: FLY012.csv, FLY012.kml

2026-06-15 15:21:22 datcon v4.3.0 rc=n/a
cmd: DatCon.4.3.0.exe FLY013.DAT
out: FLY013.csv, FLY013.kml

2026-06-15 15:21:23 datcon v4.3.0 rc=n/a
cmd: DatCon.4.3.0.exe FLY014.DAT
out: FLY014.csv, FLY014.kml

2026-06-15 15:21:24 datcon v4.3.0 rc=n/a
cmd: DatCon.4.3.0.exe FLY015.DAT
out: FLY015.csv, FLY015.kml

2026-06-15 15:21:25 datcon v4.3.0 rc=n/a
cmd: DatCon.4.3.0.exe FLY016.DAT
out: FLY016.csv, FLY016.kml

2026-06-15 15:21:25 datcon v4.3.0 rc=n/a
cmd: DatCon.4.3.0.exe FLY017.DAT
out: FLY017.csv, FLY017.kml

2026-06-15 15:21:26 datcon v4.3.0 rc=n/a
cmd: DatCon.4.3.0.exe FLY018.DAT
out: FLY018.csv, FLY018.kml

2026-06-15 15:21:27 datcon v4.3.0 rc=n/a
cmd: DatCon.4.3.0.exe FLY019.DAT
out: FLY019.csv, FLY019.kml

2026-06-15 15:21:28 datcon v4.3.0 rc=n/a
cmd: DatCon.4.3.0.exe FLY020.DAT
out: FLY020.csv, FLY020.kml

2026-06-15 15:21:29 datcon v4.3.0 rc=n/a
cmd: DatCon.4.3.0.exe FLY021.DAT
out: FLY021.csv, FLY021.kml

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap >= 60 s AND >= 90% of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs <= 25 m with >= 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a +-2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.